CHAPTER 7

The Generalized Single-Particle Model (HFB Theory)

7.1 Introduction

As we have seen in the last two chapters, many properties of nuclei can be described in terms of a model of independent particles moving in an average potential whose space dependence closely follows the matter distribution. With unfilled shells, we find additional correlations between these particles. In the BCS model we have learned to treat these correlations in a generalized single-particle picture by introducing quasi-particles and a new type of field, the "pairing potential."

The Hartree–Fock–Bogoliubov (HFB) theory generalizes and unifies both methods. Within this theory we look for the most general product wave functions consisting of independently moving quasi-particles. They are determined by a variational principle and take into account as many correlations as possible staying within a static single-particle picture. It turns out that within this approximation, the Hamiltonian reduces to two average potentials, the self-consistent field Γ, which we already know from the Hartree–Fock theory, and an additional pairing field Δ, known from the BCS theory. The field Γ contains all the long range *ph*-correlations which eventually lead to a deformed ground state (phase transition). On the other hand, Δ sums up the short-range pairing correlations that can lead to a phase transition and a superfluid state. HFB theory now mixes both descriptions and handles their interdependence. It is therefore capa-

ble of simultaneously describing all kinds of phase transition within the mean field approximation.

Since the HFB theory is a combination of both Hartree–Fock and BCS theory, many of the relevant problems have already been discussed in the foregoing chapters. We can therefore stay rather formal in deriving the necessary formulae appearing at the beginning of this chapter. In Section 7.2 we introduce a general quasi-particle picture. The nuclear wave function $|\Phi\rangle$ under consideration is defined as a vacuum of suitable quasi-particle operators, and its formal properties are discussed. Many of the formulae and some important theorems in this context are proved in Appendix E. In Section 7.3 we derive the HFB equations that determine the wave function $|\Phi\rangle$ and present several methods for solving them. The pairing-plus-quadrupole model is introduced in Section 7.4 as an example, which shows many important features of realistic calculations. Several applications of the theory are discussed in the last sections, namely, calculations of the ground state properties of deformed nuclei constrained Hartree–Fock theory (CHF) for the investigation of energy surfaces in the context of fission and the calculation of rotational spectra within the self-consistent cranking (SCC) model.

7.2 The General Bogoliubov Transformation

7.2.1 Quasi-particle Operators

The basic idea of any quasi-particle concept is to represent the ground state of a nucleus as a *vacuum* with respect to *quasi-particles*, which are defined by the low-lying excitations of neighboring nuclei. This is precisely the concept of Landau and Migdal [La 59, Mi 67], who defined the vacuum and the quasi-particles in terms of exact eigenstates of the many-body system. This theory of Fermi liquids is therefore, in principle, an exact one; however, it has the disadvantage that there is no simple mathematical relationship between these Landau–Migdal quasi-particles and the "bare" particles of the system (given by some basic operators c_l^+, c_l which may, for instance, be plane waves or harmonic oscillator states).

In this chapter, we use the so-called Bogoliubov quasi-particles [Bo 58, Bo 59a+b, BS 59, Va 61] which have a linear connection to the bare particles. They are easy to handle, but the corresponding vacuum $|\Phi\rangle$ and the one-quasi-particle states are now only approximations of the exact eigenfunctions of the many-body Hamiltonian.

We have already defined quasi-particles of a very special type within the BCS model (Eq. 6.67). In the limit of vanishing pairing correlations ($u_k, v_k = 0$), they are either particles ($\epsilon_k > \epsilon_F$) or holes ($\epsilon_k \leqslant \epsilon_F$). A very natural extension of these BCS quasi-particles is given by the most general linear transformation from the particle operators c_l^+, c_l to the *quasi-particle*

operators β_k^+, β_k of the form:

$$\beta_k^+ = \sum_l U_{lk} c_l^+ + V_{lk} c_l. \tag{7.1}$$

The indices k and l both run over the whole configuration space ($k = 1, \ldots, M$). The Hermitian conjugation of Eq. (7.1) gives us the operator β_k. We therefore have a transformation of the operators $(c_1 \ldots c_M, c_1^+ \ldots c_M^+) \to (\beta_1 \ldots \beta_M, \beta_1^+ \ldots \beta_M^+)$ which acts in a $2M$-dimensional space*

$$\begin{pmatrix} \beta \\ \beta^+ \end{pmatrix} = \begin{pmatrix} U^+ & V^+ \\ V^T & U^T \end{pmatrix} \begin{pmatrix} c \\ c^+ \end{pmatrix} = \mathfrak{W}^+ \begin{pmatrix} c \\ c^+ \end{pmatrix} \tag{7.2}$$

and which is represented by the matrix $\mathfrak{W}$:

$$\mathfrak{W} = \begin{pmatrix} U & V^* \\ V & U^* \end{pmatrix}. \tag{7.3}$$

The coefficients U, V are not completely arbitrary. We require the new operators β_k, β_k^+ to obey the same fermion commutation relations as the old ones. This restricts the matrix $\mathfrak{W}$ to being unitary.

$$\mathfrak{W}^+ \mathfrak{W} = 1 \qquad \text{and} \qquad \mathfrak{W} \mathfrak{W}^+ = 1, \tag{7.4}$$

or

$$\begin{aligned} U^+U + V^+V &= 1, & UU^+ + V^*V^T &= 1, \\ U^TV + V^TU &= 0, & UV^+ + V^*U^T &= 0, \end{aligned} \tag{7.5}$$

and allows us to invert Eq. (7.1):

$$c_l^+ = \sum_k U_{lk}^* \beta_k^+ + V_{lk} \beta_k. \tag{7.6}$$

There is a famous *theorem of Bloch and Messiah* [Zu 62, BM 62] which states that a unitary matrix $\mathfrak{W}$ of the form (7.3) can always be decomposed into three matrices of very special form:

$$\mathfrak{W} = \begin{pmatrix} D & 0 \\ 0 & D^* \end{pmatrix} \begin{pmatrix} \bar{U} & \bar{V} \\ \bar{V} & \bar{U} \end{pmatrix} \begin{pmatrix} C & 0 \\ 0 & C^* \end{pmatrix} \tag{7.7}$$

or

$$U = D\bar{U}C, \qquad V = D^*\bar{V}C. \tag{7.8}$$

*The definition of the matrix $\mathfrak{W}$ in Eq. (7.2) seems somewhat unusual; it is, however, consistent with the definition (7.1) and has the advantage that we can use the rules of matrix multiplication in the following.

D and C are unitary matrices and $\overline{U}, \overline{V}$ are real matrices of the general form

$$\overline{U} = \begin{bmatrix} 0 & & & & & & & & \\ & \ddots & & & & & & & \\ & & 0 & & & & & & \\ & & & \boxed{\begin{matrix} u_1 & 0 \\ 0 & u_1 \end{matrix}} & & & & & \\ & & & & \ddots & & & & \\ & & & & & \boxed{\begin{matrix} u_n & 0 \\ 0 & u_n \end{matrix}} & & & \\ & & & & & & 1 & & \\ & & 0 & & & & & \ddots & \\ & & & & & & & & 1 \end{bmatrix}, \tag{7.9}$$

$$\overline{V} = \begin{bmatrix} 1 & & & & & & & & \\ & \ddots & & & & & & & \\ & & 1 & & & & & & \\ & & & \boxed{\begin{matrix} 0 & v_1 \\ -v_1 & 0 \end{matrix}} & & & & & \\ & & & & \ddots & & & & \\ & & & & & \boxed{\begin{matrix} 0 & v_n \\ -v_n & 0 \end{matrix}} & & & \\ & & & & & & 0 & & \\ & 0 & & & & & & \ddots & \\ & & & & & & & & 0 \end{bmatrix}$$

The proof of this theorem is given in Section E.1. Its meaning is that the transformation (7.1) can be decomposed into three parts

$$\left.\begin{matrix} c \to a \\ c^+ \to a^+ \end{matrix}\right\} \to \left\{\begin{matrix} \alpha \to \beta \\ \alpha^+ \to \beta^+ \end{matrix}\right. \tag{7.10}$$

$$D \qquad \overline{U}, \overline{V} \qquad C$$

(i) a *unitary transformation of the particle operators* c^+ among themselves

as in the HF case [Eq. (5.18)]

$$a_k^+ = \sum_l D_{lk} c_l^+ . \tag{7.11}$$

It defines a new basis, called the "*canonical*" basis,* which is explained in more detail in Appendix E. It is convenient to use because, as we shall see, the density matrix ρ is diagonal in this basis;

(ii) a *special Bogoliubov transformation*, which distinguishes between "*paired*" levels ($u_p > 0, v_p > 0$)

$$\begin{aligned} \alpha_p^+ &= u_p a_p^+ - v_p a_{\bar{p}}, \\ \alpha_{\bar{p}}^+ &= u_p a_{\bar{p}}^+ + v_p a_p, \end{aligned} \tag{7.12}$$

where $(p, \bar{p})$ are defined by the 2×2 boxes in Eq. (7.9) (see also Chap. 6), and "*blocked*" levels, which are either occupied ($v_i = 1$; $u_i = 0$) or empty ($v_m = 0$; $u_m = 1$):

$$\begin{aligned} \alpha_i^+ &= a_i, \qquad \alpha_m^+ = a_m^+, \\ \alpha_i &= a_i^+, \qquad \alpha_m = a_m. \end{aligned} \tag{7.13}$$

(7.12) corresponds to the BCS-transformation (6.67). The orthogonality relations (7.5) guarantee that the real occupation numbers v_p and u_p are normalized;

(iii) a *unitary transformation of the quasi-particle operators* α_k^+ among themselves

$$\beta_k^+ = \sum_{k'} C_{k'k} \alpha_{k'}^+ . \tag{7.14}$$

From this theorem we see that a general Bogoliubov transformation (7.1) is nothing but a BCS transformation in an appropriate basis, the canonical basis, defined by the transformation D in Eq. (7.11). In particular, the decomposition (7.7) defines fully occupied levels (i), completely empty levels (m), and paired levels (p) with canonical conjugate states $p, \bar{p}$. In many problems with time reversal symmetry it turns out that one can choose the time reversal operation as canonical conjugation. However, there are cases (for example, the HFB-theory in a rotating frame; see Sec. 7.7) where we do not know a priori what the canonical conjugation is. First we have to determine the full HFB transformatiom (7.1) and afterwards we can apply the Bloch–Messiah theorem (7.7) for the calculation of the canonical basis. We will see in Sec. 7.2.2, how this decomposition can be achieved in practical cases, which will be useful for a deeper insight into the physical content of the wave function.

* It has this name because the skew symmetric pairing tensor κ (7.24) is in the canonical form in this basis.

7.2.2 The Quasi-particle Vacuum

The definition of quasi-particle operators in the last section has up to now been highly mathematical. To get physical information from the transformation, we have to calculate a wave function. As discussed above, the ground state of the many-body system $|\Phi\rangle$ shall be represented as the vacuum with respect to these quasi-particles:

$$\beta_k|\Phi\rangle=0 \qquad \text{for all} \quad k=1,\ldots,M. \tag{7.15}$$

Wave functions which fulfill this condition for a corresponding set of quasi-particle operators (7.1) will, in the following, be called HFB wave functions.

Before we can study the structure of such wave functions in more detail, we want to show that this definition is always possible and unique: It is easy to construct a wave function which fulfills Eq. (7.15). We start with the bare vacuum $|-\rangle$ and multiply it by a product of annihilation operators β_k:

$$|\Phi\rangle=\prod_k \beta_k|-\rangle. \tag{7.16}$$

If k runs over all values $k=1,\ldots,M$, the condition (7.15) is certainly fulfilled. In many cases, however, such a function vanishes identically. In a HF state, for instance, the product can run only over the annihilation operators of all hole states

$$|\Phi_{\mathrm{HF}}\rangle=\prod_i \alpha_i|-\rangle=\prod_i a_i^+|-\rangle. \tag{7.17}$$

Any annihilation operator of a particle state $\alpha_m=a_m$ would make $|\Phi\rangle$ vanish identically. Therefore, in Eq. (7.16) we define $\prod$ as the product of the maximal number of k-values, such that the multiplication of any additional operator β_q would annihilate $|\Phi\rangle$. This maximal number turns out to depend on the physical situation. It is determined, as we shall see later on, by the blocking structure (7.18). It may happen that this number is even; then the wave function $|\Phi\rangle$ describes an even nucleus (ground state or excited state; see below). If this number is odd, one deals with an odd nucleus. From this definition, we can always construct a function $|\Phi\rangle$ which fulfills Eq. (7.15). Since the basis sets $c_{l_1}^+,\ldots,c_{l_N}^+|-\rangle$ $(N=0,\ldots,M)$ and $\beta_{k_1}^+,\ldots,\beta_{k_N}^+|\Phi\rangle$ $(N=0,\ldots,M)$ are both orthogonal complete sets in the many-body Hilbert space, the definition of $|\Phi\rangle$ by the operators β_k $(k=1,\ldots,M)$ in Eq. (7.15) is certainly unique.

The opposite is not true: $|\Phi\rangle$ does not uniquely define the quasi-particle operators β_k. Any transformation of these operators amongst themselves [as in the C-transformation in Eq. (7.14)] leaves $|\Phi\rangle$ invariant. This means, in particular, that we could have also used the quasi-particle operators α_k [Eqs. (7.12) and (7.13)] in the canonical basis for the definition of $|\Phi\rangle$.

Therefore, $|\Phi\rangle$ is fully determined by the single-particle operators a_k in the canonical basis, and the occupation probabilities v_k^2.

In particular, we can now use the prescription (7.16) to construct a more explicit form of $|\Phi\rangle$. It is clear that we have to leave out any operator α of unoccupied levels in Eq. (7.13). After proper normalization, we find

$$|\Phi\rangle = \mathcal{N} \prod_k \alpha_k |-\rangle = \prod_i a_i^+ \prod_p (u_p + v_p a_p^+ a_{\bar{p}}^+)|-\rangle, \tag{7.18}$$

where, by definition, none of the numbers u_p vanishes. Depending on whether the number of occupied levels i is even or odd, the function (7.18) is a superposition of states with an even or odd particle number. We call this quantum number the *number parity* [BMR 73]. It is evident that a wave function $|\Phi\rangle$ with even number parity can only describe a system with even particle number, and vice versa.

We usually assume that $|\Phi\rangle$ describes the ground state of an even system, but we have not used this fact in the derivation. We now see that, dependent on the coefficients U and V in the Bogoliubov transformation (7.1) which completely determine our wave function, we get even or odd number parity. These coefficients are determined by a variation of the energy expectation value and we will derive the corresponding (HFB) equations in Section 7.3.1. They are nonlinear and can have several solutions. Usually, the solution of these equations with the deepest energy will provide us with such coefficients U and V that the corresponding vacuum $|\Phi\rangle$ describes the ground state of an even nucleus, that is, the state (7.18) can be represented as a BCS state (6.31) in the canonical basis. Sometimes, however, we want to calculate the ground state of an odd mass nucleus. In such a case, we have to make sure that we use coefficients U and V which guarantee odd number parity for the wave function $|\Phi_1\rangle$; that is, $|\Phi_1\rangle$ can be written as a one quasi-particle state based on a properly chosen ground state $|\Phi_0\rangle$ with even number parity.

In practice, this is very simple to accomplish [BMR 73, Ma 75a]. We start, for instance, with a fully paired vacuum $|\Phi_0\rangle = \beta_1 \beta_2 \ldots \beta_M |-\rangle$ with even number parity. The one-quasi-particle state

$$|\Phi_1\rangle = \beta_1^+ |\Phi_0\rangle \tag{7.19}$$

is a vacuum to the operators $(\tilde{\beta}_1, \tilde{\beta}_2, \ldots, \tilde{\beta}_M)$ with

$$\tilde{\beta}_1 = \beta_1^+, \ \tilde{\beta}_2 = \beta_2, \ldots, \tilde{\beta}_M = \beta_M. \tag{7.20}$$

The exchange of a quasi-particle creation operator β_1^+ with the corresponding annihilation operator β_1 means that we have replaced columns 1 in the matrices U and V by the corresponding columns in the matrices V^*, U^*:

$$(U_{l1}, V_{l1}) \leftrightarrow (V_{l1}^*, U_{l1}^*). \tag{7.21}$$

Thus by making such a replacement we change the number parity of the corresponding vacuum and go over to a one-quasi-particle state. This can be continued—starting from the fully paired ground state we can come to

many-quasi-particle states by simply interchanging the corresponding columns in the HFB coefficients. With this trick we represent *quasi-particle excitations* as HFB *vacua* for properly defined new quasi-particle operators.

Of course, the transformations C, $\bar{U}$, $\bar{V}$, and D of Eq. (7.7) are changed by the replacement (7.20). In particular, the canonical basis for $|\Phi_1\rangle$ is, in general, different from that for $|\Phi_0\rangle$ in Eq. (7.18). Only in cases where the C transformation is equal to unity (i.e., if $|\Phi_1\rangle = \alpha_1|\Phi_0\rangle$) do the two wave functions have the same canonical basis.

7.2.3 The Density Matrix and the Pairing Tensor

As we saw in the last section, the general Bogoliubov transformation (7.1) or the coefficients U_{lk} and V_{lk} are not uniquely defined by the HFB wave function $|\Phi\rangle$ to within a C transformation (7.14).

We now define two quantities which contain no redundant information and which determine the wave function $|\Phi\rangle$ uniquely. They are called the *normal* and *abnormal density* (or density matrix and pairing tensor), and are given by their matrix elements in the particle basis

$$\rho_{ll'} = \langle\Phi|c_{l'}^{+}c_l|\Phi\rangle, \qquad \kappa_{ll'} = \langle\Phi|c_{l'}c_l|\Phi\rangle, \tag{7.22}$$

or in matrix notation

$$\rho = V^{*}V^{T}, \qquad \kappa = V^{*}U^{T} = -UV^{+}. \tag{7.23}$$

ρ is hermitian ($\rho^{+} = \rho$) and κ is skew symmetric ($\kappa^{T} = -\kappa$).

Using the decomposition of the Bloch–Messiah theorem [Eq. (7.9)] and the unitarity of C, we find

$$\rho = D\bar{V}^{2}D^{+}, \qquad \kappa = D\bar{U}\bar{V}D^{T}. \tag{7.24}$$

This means that ρ is diagonal in the canonical basis. The eigenvalues of ρ are the occupation probabilities v_k^2 and the eigenvectors are the coefficients D_{lk} of the wave functions a_k (7.11) in the canonical basis.* At the same time, κ is in its canonical form: it decomposes into 2×2 matrices:

$$\begin{pmatrix} 0 & u_k v_k \\ -u_k v_k & 0 \end{pmatrix}. \tag{7.25}$$

Two important relations hold for ρ and κ. They follow from Eqs. (7.5) and (7.23):

$$\rho^{2} - \rho = -\kappa\kappa^{+}, \qquad \rho\kappa = \kappa\rho^{*}. \tag{7.26}$$

Sometimes it is useful to define a $2M$-dimensional generalized density

* Equation (7.24) shows that ρ behaves under a unitary transformation like a linear operator and κ behaves like the matrix part of an antilinear operator. For the connection between HFB theory and the theory of antilinear operators, see [HV 68, VH 70].

matrix [Sch 61, Va 61]

$$\mathfrak{R}=\begin{pmatrix}\langle\Phi|\, c_{l'}^{+}c_{l}\,|\Phi\rangle & \langle\Phi|\, c_{l'}c_{l}\,|\Phi\rangle\\ \langle\Phi|c_{l'}^{+}c_{l}^{+}|\Phi\rangle & \langle\Phi|c_{l'}c_{l}^{+}|\Phi\rangle\end{pmatrix}=\begin{pmatrix}\rho & \kappa\\ -\kappa^{*} & 1-\rho^{*}\end{pmatrix}, \tag{7.27}$$

which is Hermitian and idempotent:

$$\mathfrak{R}^{2}=\mathfrak{R}. \tag{7.28}$$

Its eigenvectors are the HFB coefficients $\binom{U}{V}$ for quasi-particle creation operators with eigenvalue 0, and $\binom{V^{*}}{U^{*}}$ for quasi-particle annihilation operators with eigenvalue 1:

$$\mathfrak{W}^{+}\mathfrak{R}\mathfrak{W}=\begin{pmatrix}\langle\Phi|\,\beta_{k'}^{+}\beta_{k}\,|\Phi\rangle & \langle\Phi|\,\beta_{l'}\beta_{l}\,|\Phi\rangle\\ \langle\Phi|\,\beta_{k'}^{+}\beta_{k}^{+}|\Phi\rangle & \langle\Phi|\,\beta_{l'}\beta_{l}^{+}|\Phi\rangle\end{pmatrix}=\begin{pmatrix}0 & 0\\ 0 & 1\end{pmatrix}. \tag{7.29}$$

Both sets of eigenvectors are determined only up to a unitary transformation in the corresponding eigenspace, that is, up to the C transformation (7.14), and this shows that ρ and κ uniquely determine $|\Phi\rangle$.

In the following chapters we will see that many theories developed originally in the HF picture of pure Slater determinants ($\rho^{2}=\rho$) can be immediately generalized to the HFB case with pairing correlations simply by working in the $2M$-dimensional formalism with the matrix $\mathfrak{R}$.

7.3 The Hartree–Fock–Bogoliubov Equations

7.3.1 Derivation of the HFB Equation

Until now we have investigated only the formal structure of the HFB transformation (7.1) and the corresponding vacuum $|\Phi\rangle$. In this section we derive an equation for the coefficients U_{lk} and V_{lk} which defines the quasi-particles and the wave function $|\Phi\rangle$.

We assume that $|\Phi\rangle$ is an approximation for the exact ground state of the Hamiltonian

$$H=\sum_{l_1l_2}t_{l_1l_2}c_{l_1}^{+}c_{l_2}+\frac{1}{4}\sum_{l_1l_2l_3l_4}\bar{v}_{l_1l_2l_3l_4}c_{l_1}^{+}c_{l_2}^{+}c_{l_4}c_{l_3} \tag{7.30}$$

and use the variation principle of Ritz (see Sec. 5.2). In our case, the trial wave functions are the set of all generalized product states $|\Phi\rangle$ of the HFB type.

As in the BCS model these wave functions violate the symmetry connected with the particle number (see Sec. 6.3). Again we have to use a constraint on the particle number N and vary the Hamiltonian $H'=H-\lambda N$. To simplify the notation, we neglect the prime in the following and come back to the problem of a variation with constraint in Section 7.6.

Starting from the variational principle

$$\delta\frac{\langle\Phi|H|\Phi\rangle}{\langle\Phi\,|\,\Phi\rangle}=0, \tag{7.31}$$

we have to investigate small variations $|\delta\Phi\rangle$ in the vicinity of the solution $|\Phi\rangle$. Since we consider only wave functions of the HFB type, we can use a theorem of Thouless (see Appendix E) and express the function $|\Phi'\rangle = |\Phi\rangle + |\delta\Phi\rangle$, which is not orthogonal to $|\Phi\rangle$ [MW 68, Ma 75a] by:

$$|\Phi'\rangle = \exp\Big(\sum_{k<k'} Z_{kk'}\beta_k^+\beta_{k'}^+\Big)|\Phi\rangle. \tag{7.32}$$

In contrast to the coefficients U_{lk} and V_{lk}, which obey orthogonality relations (7.5), the variables $Z_{kk'}$ (with $k<k'$) are independent variables. The solution $|\Phi\rangle$ of variational equation (7.31) corresponds to $Z_{kk'}=0$. For infinitesimal variations, we can expand up to second order. Using quasi-particle representation (E.18) for the Hamiltonian

$$H = H^0 + \sum_{k_1k_2} H^{11}_{k_1k_2}\beta^+_{k_1}\beta_{k_2} + \sum_{k_1<k_2}\big(H^{20}_{k_1k_2}\beta^+_{k_1}\beta^+_{k_2} + \text{h.c.}\big) + H_{\text{int}}, \tag{7.33}$$

we get

$$\frac{\langle\Phi'|H|\Phi'\rangle}{\langle\Phi'|\Phi'\rangle} = H^0 + (H^{20*}H^{20})\begin{pmatrix} Z \\ Z^* \end{pmatrix} + \frac{1}{2}(Z^*Z)\begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix}\begin{pmatrix} Z \\ Z^* \end{pmatrix}, \tag{7.34}$$

where the index of the vectors and matrices runs over all pairs $(k<k')$ and

$$\begin{aligned} H^0 &= \langle\Phi|H|\Phi\rangle, & A_{kk'll'} &= \langle\Phi|\big[\beta_{k'}\beta_k, \big[H, \beta_l^+\beta_{l'}^+\big]\big]|\Phi\rangle; \\ H^{20}_{kk'} &= \langle\Phi|\big[\beta_{k'}\beta_k, H\big]|\Phi\rangle, & B_{kk'll'} &= -\langle\Phi|\big[\beta_{k'}\beta_k, \big[H, \beta_{l'}\beta_l\big]\big]|\Phi\rangle. \end{aligned} \tag{7.35}$$

Equation (7.34) gives a quadratic approximation of the multidimensional energy surface in the vicinity of $|\Phi\rangle$. The variation with respect to $Z^*_{kk'}$ yields* [Be 59]

$$\frac{\partial}{\partial Z^*_{kk'}}\frac{\langle\Phi'|H|\Phi'\rangle}{\langle\Phi'|\Phi'\rangle}\bigg|_{Z=0} = H^{20}_{kk'} = 0, \tag{7.36}$$

which means that the linear terms in Eq. (7.34) vanish at the stationary point. To see if it is a minimum or a saddle point, the quadratic terms in Eq. (7.34) must be investigated. The matrix

$$\mathcal{S} = \begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix} \tag{7.37}$$

is called the *stability matrix* (or *curvature tensor*). At a minimum it has to be positive definite.

The variational equations (7.36) are not affected by a C-transformation of the quasi-particles among themselves [Eq. (7.14)]. The requirement $H^{20}=0$ determines, therefore, only the first two of the Bloch–Messiah transformations. The third transformation can be used to diagonalize the H^{11} part in Eq. (7.33). Together with (7.36), this corresponds to the

* Variation with respect to $Z_{kk'}$ gives the complex conjugate equation.

diagonalization of the supermatrix

$$\begin{pmatrix} H^{11} & H^{20} \\ -H^{20*} & -H^{11*} \end{pmatrix} = \begin{pmatrix} \langle\Phi|\{[\beta_k, H], \beta_{k'}^+\}|\Phi\rangle & \langle\Phi|\{[\beta_k, H], \beta_{k'}\}|\Phi\rangle \\ \langle\Phi|\{[\beta_k^+, H], \beta_{k'}^+\}|\Phi\rangle & \langle\Phi|\{[\beta_k^+, H], \beta_{k'}\}|\Phi\rangle \end{pmatrix}. \tag{7.38}$$

In the space of the basis operators c_l, c_l^+ this matrix has the form

$$\mathcal{H} = \mathcal{W}\begin{pmatrix} H^{11} & H^{20} \\ -H^{20*} & -H^{11*} \end{pmatrix}\mathcal{W}^+ = \begin{pmatrix} h & \Delta \\ -\Delta^* & -h^* \end{pmatrix} \tag{7.39}$$

with

$$h_{ll'} = \langle\Phi|\{[c_l, H], c_{l'}^+\}|\Phi\rangle, \qquad \Delta_{ll'} = \langle\Phi|\{[c_l, H], c_{l'}\}|\Phi\rangle. \tag{7.40}$$

Applying Wick's theorem (Sec. C.4), we get*

$$h = \epsilon + \Gamma - \lambda; \qquad \Gamma_{ll'} = \sum_{qq'} \bar{v}_{lq'l'q}\rho_{qq'}; \qquad \Delta_{ll'} = \frac{1}{2}\sum_{qq'} \bar{v}_{ll'qq'}\kappa_{qq'}. \tag{7.41}$$

We therefore end up with a diagonalization problem for the matrix $\mathcal{H}$, the so-called HFB *equations* [Ba 61, 63a+b]:

$$\begin{pmatrix} h & \Delta \\ -\Delta^* & -h^* \end{pmatrix}\begin{pmatrix} U_k \\ V_k \end{pmatrix} = \begin{pmatrix} U_k \\ V_k \end{pmatrix}\cdot E_k, \tag{7.42}$$

where the columns U_k, V_k of the matrices U and V determine the quasi-particle operator β_k^+ (7.1). In the basis corresponding to the operators β_k, both matrices $\mathcal{H}$ and $\mathcal{R}$ [see Eq. (7.29)] are diagonal. We therefore get as an equivalent condition to Eq. (7.36) [BS 59]:

$$[\mathcal{H}, \mathcal{R}] = 0 \tag{7.43}$$

which corresponds exactly to the formulation (5.36) of the HF equation.

The Hamiltonian (7.33) now takes the form

$$H = H^0 + \sum_k E_k\beta_k^+\beta_k + H_{\text{int}}. \tag{7.44}$$

H_{int} (E.18) contains the terms H^{40}, H^{31}, and H^{22}. They are neglected in the HFB approach. In this case, H is diagonal. Its eigenstates are the quasi-particle vacuum $|\Phi\rangle$ (with the eigenvalue H^0), one-quasi-particle states

$$|\Phi_k\rangle = \beta_k^+|\Phi\rangle, \tag{7.45}$$

with the quasi-particle energies E_k, two-quasi-particles states, and so on. The excited states to $|\Phi\rangle$ are states with an even number of quasi-particles. The states with an odd number of quasi-particles describe the neighboring nuclei ($A \pm 1$).

These considerations, however, give only a rough overview of the structure of excited states. In particular, the one-quasi-particle states (7.45) are not determined self-consistently. We will see in the next section how this point can be improved.

*The chemical potential λ is determined by the particle number: $\text{Tr}\,\rho = N$.

Before we discuss the properties of the HFB equations (7.42), let us give a different derivation which shows some interesting aspects of the theory and requires less calculation [Bl 62].

Using the theorem of Wick (see Sec. C.4), we expand the Hamiltonian in normal order (: :) with respect to the ground state $|\Phi\rangle$. We then get for the one-particle operator

$$\begin{aligned} c_l^+ c_{l'} &= \langle\Phi|c_l^+ c_{l'}|\Phi\rangle + : c_l^+ c_{l'} : \\ &= \rho_{l'l} + : c_l^+ c_{l'} :, \end{aligned} \tag{7.46}$$

and for the two-particle operator

$$\begin{aligned} c_{l_1}^+ c_{l_2}^+ c_{l_4} c_{l_3} = {} & \rho_{l_3 l_4}\rho_{l_4 l_2} - \rho_{l_3 l_2}\rho_{l_4 l_1} + \kappa^*_{l_1 l_2}\kappa_{l_3 l_4} \\ & + \rho_{l_3 l_1} : c_{l_2}^+ c_{l_4} : + \rho_{l_4 l_2} : c_{l_1}^+ c_{l_3} : - (l_3 \leftrightarrow l_4) \\ & + \kappa^*_{l_1 l_2} : c_{l_4} c_{l_3} : + \kappa_{l_3 l_4} : c_{l_1}^+ c_{l_2}^+ : \\ & + : c_{l_1}^+ c_{l_2}^+ c_{l_4} c_{l_3} : . \end{aligned} \tag{7.47}$$

Using the definitions (7.41) for Γ and Δ, we immediately find

$$H = H^0 + \frac{1}{2} : (c^+ c) \begin{pmatrix} h & \Delta \\ -\Delta^* & -h^* \end{pmatrix} \begin{pmatrix} c \\ c^+ \end{pmatrix} : + \frac{1}{4} \sum_{l_1 l_2 l_3 l_4} \bar{v}_{l_1 l_2 l_3 l_4} : c_{l_1}^+ c_{l_2}^+ c_{l_4} c_{l_3} : . \tag{7.48}$$

The last term contains only products of four-quasi-particle operators. It corresponds to H^{40}, H^{31}, and H^{22} in the quasi-particle representation (E.18). In the generalized single-particle model, this interaction between the quasi-particles is neglected. The rest is a quadratic form in the operators c, c^+, which can be diagonalized exactly by the general Bogoliubov transformation (7.1). Again we find the HFB equations (7.42) with the solution (7.44).

7.3.2 Properties of the HFB Equations

The HFB-equations (7.42) are a $2M$-dimensional set of nonlinear equations. They show more or less the same properties as the HF equations (5.38). Many points of the discussion in Chapter 5 are therefore also valid for the HFB equations. In particular, we can also derive them for density dependent forces by a variation with respect to both densities, ρ and κ (see Sec. 7.5). The theorem of self-consistent symmetries (Chap. 5) applies here also [Sa 68, Go 79]. They can be solved either by iterative diagonalization of the matrix $\mathcal{H}$ in Eq. (7.39) or by the gradient method, which will be described in the next section.

There are, however, some basic differences to the HF case, which we will discuss in the following.

The equations contain a chemical potential λ which has to be determined by the particle number subsidiary condition. Therefore, we must always constrain HFB-equations; for the method of treating such problems, see Sections (7.3.3) and (7.6).

In addition, the HFB equations contain two potentials, Γ and Δ. The Γ corresponds to the HF potential, which describes the shape of the nucleus (spherical or deformed), whereas Δ determines the pairing correlations. In contrast to the BCS-theory, Δ is now no longer given by one number Δ_k [Eq. (6.50)], but is a matrix which mixes different levels. Since we have seen that the HFB wave functions $|\Phi\rangle$ have the structure of BCS wave functions in the canonical basis

(7.11), it is interesting to investigate the HFB equations (7.42) in this basis. We therefore use the matrices $\overline{U}$ and $\overline{V}$ (7.9), for example, in a fully paired case, and obtain from Eq. (E.22), for the diagonal matrix elements of $H_{k\bar{k}}^{20}$ in the canonical basis (for real matrix elements $\Delta_{k\bar{k}}$),

$$u_k v_k (h_{kk} + h_{\bar{k}\bar{k}}) + \Delta_{k\bar{k}} (u_k^2 - v_k^2) = 0, \tag{7.49}$$

which has a similar solution to Eq. (6.52) viz:

$$\begin{pmatrix} u_k^2 \\ v_k^2 \end{pmatrix} = \frac{1}{2} \left[1 \pm \frac{\frac{1}{2}(h_{kk} + h_{\bar{k}\bar{k}})}{\sqrt{\frac{1}{4}(h_{kk} + h_{\bar{k}\bar{k}})^2 + \Delta_{k\bar{k}}^2}} \right]. \tag{7.50}$$

We see that the important quantities, which determine the occupation numbers, are the diagonal matrix elements h_{kk}, $h_{\bar{k}\bar{k}}$, and $\Delta_{k\bar{k}}$ in the canonical basis. This does not, however, mean that the other matrix elements of h and Δ in general vanish, and a nontrivial C-transformation is needed to diagonalize H^{11}.

There is a special case when H^{11} is already diagonal in the canonical basis. As we see from Eq. (E.22) for H^{20}, this happens in cases of time reversal symmetry, whenever the only non-vanishing matrix elements of Δ in the canonical basis are the elements $\Delta_{k\bar{k}}$. In such cases, the off-diagonal matrix elements of h in the canonical basis also vanish, and we get for the quasi-particle energies

$$E_k = \sqrt{h_{kk}^2 + \Delta_{k\bar{k}}^2} \ . \tag{7.51}$$

This happens, for instance, in all cases where there is time reversal invariance together with a monopole pairing force for the calculation of the pair field Δ (see Sec. 7.4). In such cases, the full HFB equations (7.42) can be solved by a solution of HF equations (5.39), including the appropriate occupation factors v_k^2, which themselves are determined by the BCS equations in each step of the iteration. In the general case this is, however, not true: The Bloch–Messiah theorem (Sec. 7.2.1.) does not imply that HFB can be replaced by coupled HF–BCS equations (see also [Go 79]).

We want to stress that for the derivation of the HFB equations we have only used the general product structure of the function $|\Phi\rangle$ and the stationarity condition (7.31). Whether the solution corresponds to the absolute minimum in the multidimensional energy surface (see Sec. 7.3.3) or only to a local one, or perhaps only to a saddle point, depends on how we solve these equations and, in particular, on the initial conditions of the iterative solution. For instance, if we use a self-consistent symmetry (see Sec. 5.5), we can never get an eventual absolute minimum which breaks this symmetry. In such a case, the solution of the HFB equations will therefore be a local minimum or a saddle point.

This fact is often used to approximate functions other than the ground state wave function of the system. If there is a symmetry operator (for instance, angular momentum or the K-quantum number), the many-body Hamiltonian H can be diagonalized in each eigenspace of this symmetry separately, and we can apply the HFB approximation in each case. In such cases, the separate solution of the HFB equations in each subspace provides us with approximations for the lowest eigenstates with the corresponding symmetry.

An example is the number parity (see Sec. 7.2.2). If we run the iteration in such a way that $|\Phi\rangle$ has at each step an odd number parity, we get an approximation to the wave function for an odd mass nucleus. This corresponds to the self-consistent solution of the HFB equations for odd mass nuclei [RBM 70, RMB 74]. It includes,

in particular, the blocking effect (see Sec. 6.3.4) and the change of the average potential due to the odd particle (polarization). How such calculations are acheived in practice will be discussed later in this section.

Another example is two-quasi-particle states in the even system with axial symmetry. If the K-value of such a state does not vanish, it is automatically orthogonal to the ground state with $K=0$. For the calculation of $K=0$, two-quasi-particle states, we have to realize that such states correspond to different stationary points in the energy surface of the ground state. They can be found by a variation under the constraint that the wave function is orthogonal to the ground state wave function [MSR 76, EMR 80a]. Only in cases where they lie in the minimum of a well separated valley with approximately orthogonal wave functions, can we neglect this constraint.

We see that the HFB equations (7.42), in principle, can be used for the self-consistent approximation of all eigenstates of H whose wave functions have the structure of generalized products.

In an actual solution by iterative diagonalization of Eq. (7.42), we encounter the problem that the equations are $2M$-dimensional and have $2M$ eigenvalues and eigenvectors. To construct a set of quasi-particle operators $\{\beta_1^+ \ldots \beta_M^+\}$, and for the calculation of ρ and κ in Eq. (7.23), we have to choose M of them. However, as is easily recognized, to each eigenvector (V_k, U_k) with eigenvalue E_k there corresponds an eigenvector (V_k^*, U_k^*) with eigenvalue $-E_k$. We saw in Section 7.2.2 that an exchange of these two eigenvectors corresponds to a replacement of one operator β_k^+ by β_k and vice versa. It is forbidden to choose E_k and $-E_k$ at the same time (otherwise it is impossible to fulfill the Fermi commutation relations for the operators β, β^+). Therefore, we have to decide for each k ($k=1 \ldots M$) whether one takes the eigenvalue E_k with the eigenvector (U_k, V_k) or $-E_k$ with the eigenvector (V_k^*, U_k^*).

In the pure HF case, this choice corresponds to the freedom of keeping a level occupied or empty in the calculation of the density matrix ρ. Of course, in such a calculation the levels are usually filled according to their energy to get the lowest total energy. This corresponds to the case where all the excitation energies of the system are positive.

In the same way, we usually choose the M positive eigenvalues E_k in the HFB case for *even particle number*. Then the excited states of the system, whose lowest are the two-quasi-particle states, have a positive energy. With this choice at each step of the iteration we usually find a fully paired state which corresponds to the deepest minimum in the energy surface.

If we want to solve the HFB equations for an *odd nucleus*, we can start with the solution of the neighboring even system and a pure quasi-particle state as the input for the iteration. As we have already seen in Section 7.2.2, such a state can be interpreted as a "vacuum" to a new set of quasi-particle operators $\tilde{\beta}$ [see Eq. (7.20)] by interchanging one of the eigenvectors (U_k, V_k) by (V_k^*, U_k^*). We therefore have, in the solution of the HFB equation of an odd system after each diagonalization, to choose one of the negative energies $-E_k$ with the corresponding eigenvector (V_k^*, U_k^*) to get a blocked HFB function. Which value of k we take depends on the quasi-particle state we want to describe. They represent different local minima in the energy surface. By such an exchange we therefore jump from one valley to another.

In cases where such valleys are no longer well separated (for instance, in band crossing phenomena in even nuclei), it may happen that one of the quasi-particle energies becomes very small and even *negative* (see Fig. 7.8d). In such cases we

must even use a negative quasi-particle energy for the even system. We are not allowed to choose only positive ones, because the latter would correspond to a jump into a different valley with odd number parity. Consequently, we must investigate the blocking structure of the system in each step of the iteration [BMR 73]. In the next section we will study a method that avoids such problems.

7.3.3 The Gradient Method

The HFB equations (7.42) are often solved by iterative diagonalization. In the last subsection we discussed the problems that can arise in such a method. The procedure is often rather time-consuming and convergence is not always guaranteed (see, for instance, Sec. 5.4), and it gets particularly complicated in cases with one or several subsidiary conditions.

In the following we will present the gradient method for the solution of the HF or the HFB equations [MSR 76, EMR 80a]. It is based on the fact that we are interested in a local minimum of the multidimensional energy surface.* Starting from an arbitrary initial wave function $|\Phi_0\rangle$, which corresponds to one point on this surface, it searches for the direction of steepest descent and follows it in iterative steps until the minimum is reached.

What we need, therefore, is a suitable and unique parametrization of the energy surface, at least in the vicinity of the point $|\Phi_0\rangle$. We use the theorem of Thouless (see Sec. E.3), which states that each wave function of the HFB type which is not orthogonal to $|\Phi_0\rangle$ can be represented as

$$|\Phi(Z)\rangle \propto \exp\Big(\sum_{k<k'} Z_{kk'}\beta_k^+\beta_{k'}^+\Big)|\Phi_0\rangle, \tag{7.52}$$

where the operators β_k annihilate $|\Phi_0\rangle$ and there is a one-to-one correspondence between the $M(M\text{-}1)/2$ parameters $Z_{kk'}$ $(k<k')$ and the HFB functions $|\Phi(Z)\rangle$. As discussed in Sec. E.3, functions $|\Phi\rangle$ orthogonal to $|\Phi_0\rangle$ have, in a sense, the structure of quasi-particle excitations with a diverging matrix Z. They are therefore "far away" from $|\Phi_0\rangle$ on the energy surface and cannot be reached using a finite Z. Within this method it is therefore impossible to jump in one step to a state which is orthogonal to $|\Phi_0\rangle$, as is sometimes done in the diagonalization method. Nevertheless it is possible to reach all HFB wave functions in several steps, because at each point of the iteration the Thouless theorem is again applied. This corresponds to a new parametrization of the energy surface at each step.

Starting from $|\Phi_0\rangle$ (given, for instance, by the HFB coefficients U_0, V_0), we go as a first step to a function $|\Phi_1\rangle$ which corresponds to the parameters Z_1 chosen in the direction of the steepest descent, that is, parallel to the gradient of the energy [see Eq. (7.36)]

$$Z_{1kk'} = -\eta \frac{\partial}{\partial Z_{kk'}^*} E(Z)\Big|_{Z=0} = -\eta H_{kk'}^{20}. \tag{7.53}$$

The parameter η determines the size of the step. It is somewhat arbitrary at the beginning and reduced or increased in the following iterations, depending on the actual values of the energies $E(Z_i)$. To get the new coefficients U_1, V_1 that

*See also [Mc 56, 60, BTR 77].

correspond to the function $|\Phi_1\rangle = |\Phi(Z_1)\rangle$, we use Eq. (E.30) and calculate

$$\begin{aligned} U_1 &= U_0 + V_0^* Z_1^*, \\ V_1 &= V_0 + U_0^* Z_1^*. \end{aligned} \tag{7.54}$$

For general values of $Z_{1kk'}$ the coefficients U_1 and V_1 correspond to quasi-particle operators $\gamma_{k'}$ which annihilate $|\Phi(Z_1)\rangle$ but do not fulfil Fermi commutation relations

$$\{\gamma_k, \gamma_{k'}\} = 0; \qquad \{\gamma_k, \gamma_{k'}^+\} = \delta_{kk'} - (Z_1 Z_1^*)_{kk'}. \tag{7.55}$$

Therefore, we have to orthogonalize the vectors $\binom{U_1}{V_1}$ by a C-transformation (7.14) in each step of the iteration.

This procedure is continued until convergence is achieved, that is, until the gradient H^{20} vanishes identically, which corresponds to the condition (7.36).

The method is extremely useful in cases where we must fulfill a subsidiary condition—for instance, the particle number condition $\langle \hat{N} \rangle = N$. Starting from $|\Phi_0\rangle$ with arbitrary particle number N_0, we do not proceed in the direction of the gradient H^{20} alone, but we admix the gradient of the particle number N^{20}

$$Z_1 = -\eta(H^{20} - \lambda N^{20}). \tag{7.56}$$

The parameter λ is determined in such a way that $\Phi(Z_1)$ has the right particle number N up to linear order in Z_1. This gives

$$N - N_0 = \sum_{k<k'} Z_{1kk'}^* N_{kk'}^{20} + \text{c.c.} = Z_1 \cdot N^{20} = -\eta(H^{20} \cdot N^{20} - \lambda N^{20} \cdot N^{20}) \tag{7.57}$$

or

$$\lambda = \frac{H^{20} \cdot N^{20}}{N^{20} \cdot N^{20}} + \frac{N - N_0}{\eta \cdot N^{20} \cdot N^{20}}, \tag{7.58}$$

where $Z \cdot N$ is the scalar product of the vectors (Z, Z^*) and (N, N^*). In cases where $|\Phi_0\rangle$ already has the right particle number N, we get

$$Z_1 = -\eta\left(H^{20} - \frac{H^{20} \cdot N^{20}}{N^{20} \cdot N^{20}} N^{20}\right), \tag{7.59}$$

that is, we choose the direction of the gradient projected onto the hypersurface orthogonal to the gradient of N, which means that it has constant particle number. If convergence is achieved, we find, as in a variation with a Lagrange parameter λ,

$$(H - \lambda N)^{20} = 0. \tag{7.60}$$

It is easy to extend the method to several subsidiary conditions.

The path on the energy surface to choose in the gradient method can be quite different from that of the diagonalization method. An example is given in Fig. 5.3.

7.4 The Pairing-plus-Quadrupole Model

In deriving the HFB equation (7.42) we started from the Hamiltonian (7.30) and obtained the two potentials Γ and Δ [Eq. (7.41)]. Both potentials are calculated from the same nuclear interaction $v(1, 2)$. We saw in Chapter 5 that the bare nucleon–nucleon force cannot be used for the calculation of the HF potential Γ. The Brückner formalism (Chap. 4)

shows that we have to apply an effective interaction, that is, the G-matrix, which sums up part of the higher order processes. It is evident that similar arguments hold for the pairing field. Green's function techniques (see Appendix F) allow the derivation of the HFB equations, too [Go 58, AGD 63, Mi 67], and from there we see that the effective interactions for the calculation of the self-consistent field Γ and the pairing field Δ may be different. Therefore, it seems to be meaningful to also use different effective forces for both potentials in phenomenological models.

The *pairing-plus-quadrupole model* was first suggested by Bohr and Mottelson and afterwards widely used by Kisslinger and Sorenson [KS 60], Baranger and Kumar [BK 65, 68], and many other authors (for a review, see [BS 69]). It distinguishes three important effects in any mean field theory of the nuclear many-body system.

(i) The interaction between the particles can be summed up, as a first approach, by an average spherical single-particle potential which is localized in space and breaks the translational invariance.
(ii) In open shell nuclei, there are two types of additional correlations: long-range *ph*-correlations can be taken into account by a deformation of the mean field. At this point the rotational symmetry is lost.
(iii) Short-range *pp*-correlations are treated by a self-consistent pairing potential Δ which violates particle number symmetry.

These three effects are included most simply in the pairing-plus-quadrupole model. The average spherical potential is approximated by a spherical harmonic oscillator with $\mathbf{l}\cdot\mathbf{s}$ and $\mathbf{l}^2$ term, that is, by a Nilsson potential (Chap. 2) at zero deformation with corresponding single-particle energies ϵ_k. Only nucleons within one major shell of each parity feel the residual interaction. The residual interaction has two parts, one contributing to Γ (which here is only that part of the field going beyond the spherical part already contained in the single-particle energies ϵ_k), and the second which contributes to Δ.

Both parts of the interaction are chosen to be separable in the appropriate indices. Since the most import deformations are of a quadrupole nature, and since the most important pairing correlations are the ones for $I=0$ pairs, the most simple Hamiltonian of this kind has the form (see also Sec. 4.4.7):

$$H=\sum_k \epsilon_k c_k^+ c_k - \tfrac{1}{2}\chi \sum_{\mu=-2}^{2} :Q_\mu^+ Q_\mu: - GP^+P, \tag{7.61}$$

with the quadrupole operator

$$Q_\mu=\sum_{kk'}\langle k|r^2 Y_{2\mu}|k'\rangle c_k^+ c_{k'}, \tag{7.62}$$

and the creation operator for a Cooper pair

$$P^+=\sum_{k>0} c_k^+ c_{\bar{k}}^+ . \tag{7.63}$$

Here $\bar{k}$ is the time reversed state of k. BCS phases are used [see (Eq. 4.138)]. The actual values of the force constants χ and G depend on the configuration space under consideration (see, for instance, [BK 68]) and are adjusted from experimental data.

In the following, we neglect (by definition of the model) the single-particle part of $Q^+ \cdot Q$, the contributions of the pairing force to the potential Γ, and those parts of the QQ-force which contribute to the potential Δ and the Fock term in Γ. The latter terms are usually rather small. From (7.41) and (7.48) we then obtain the HFB single-particle Hamiltonian of the form:

$$H_{\mathrm{HFB}} = \sum_k \epsilon_k c_k^+ c_k - \tfrac{1}{2} \sum_{\mu=-2}^{2} q_\mu (Q_\mu^+ + Q_\mu) - p_0 (P^+ + P) \qquad (7.64)$$

with

$$q_\mu = \chi \langle \Phi | Q_\mu | \Phi \rangle \qquad \text{and} \qquad p_0 = G \langle \Phi | P | \Phi \rangle \qquad (7.65)$$

where we have assumed that the parameters q_μ and p_0 are real. The HFB Hamiltonian operator is a particle-number and rotation symmetry violating single-particle operator. With fixed "deformation" parameters q_μ and p_0 it is similar to a Nilsson Hamiltonian. In Eq. (7.65), q_μ and p_0, however, depend on the solution $|\Phi\rangle$, that is, on ρ and κ. They have to be determined by iteration, or by what is equivalent in the case of separable forces—minimizing the energy $E(q_\mu, p_0)$ with respect to those parameters.

It becomes convenient to choose the principal axis of the density distribution as the axis of our coordinate frame (this implies $q_1 = q_{-1} = 0$ and $q_2 = q_{-2}$) and to work in a basis which is symmetrized with respect to a rotation of 180° around the x-axis ([RBM 70, Go 74]; see also Eq. (1.57)):

$$|k\rangle = \frac{1}{\sqrt{2}} (|nljm\rangle - i e^{i\pi j_x} |nljm\rangle); \qquad |\bar{k}\rangle = T|k\rangle. \qquad (7.66)$$

Assuming $R_x = e^{i\pi J_x}$ as a self-consistent symmetry in the sense of Section 5.5 and arranging the levels in the order $(k_1, k_2, \ldots, \bar{k}_1, \bar{k}_2, \ldots)$ we find in general for the densities ρ and κ and for the potentials Γ and Δ, the form:

$$\rho = \begin{pmatrix} \rho_1 & 0 \\ 0 & \rho_2 \end{pmatrix}, \quad \kappa = \begin{pmatrix} 0 & \kappa_2 \\ \kappa_1 & 0 \end{pmatrix}, \quad \Gamma = \begin{pmatrix} \Gamma_1 & 0 \\ 0 & \Gamma_2 \end{pmatrix}, \quad \Delta = \begin{pmatrix} 0 & \Delta_2 \\ \Delta_1 & 0 \end{pmatrix}, \qquad (7.67)$$

with $\rho_i^+ = \rho_i$, $\Gamma_i^+ = \Gamma_i$; $\kappa_2 = -\kappa_1^T$, and $\Delta_2 = -\Delta_1^T$. The diagonalization of ρ gives the first transformation of the Bloch Messiah theorem (7.7):

$$D = \begin{pmatrix} D_1 & 0 \\ 0 & D_2 \end{pmatrix}. \qquad (7.68)$$

The Hamiltonian (7.61) is time reversal invariant, which implies

$$\begin{aligned} &\rho_2 = \rho_1^*; \quad \kappa_1^+ = \kappa_1; \quad \Gamma_2 = \Gamma_1^*; \\ &D_2 = D_1^*; \quad \Delta_1^+ = \Delta_1. \end{aligned} \qquad (7.69)$$

For a pure pairing force, since $\Delta_1 = p_0 \cdot 1$ is a multiple of unity, we find in the canonical basis $\Delta \to D^+ \Delta D^*$ or $\Delta_1 \to D_2^T \Delta_1 D_1^* = p_0 \cdot 1$, that is, Δ is invariant under the first Bloch–Messiah transformation.

To determine the transformation D it is therefore sufficient to diagonalize the self-consistent field:

$$h = \epsilon - \lambda + \Gamma = \epsilon - \lambda - q_0 Q_0 - q_2 (Q_2 + Q_{-2}). \tag{7.70}$$

This is exactly a Nilsson Hamiltonian (2.89) for fixed deformation parameters q_0, q_2. The canonical basis in this case is therefore the Nilsson basis with the eigenvalues $\tilde{\epsilon}_k$. In this basis the HFB equations (7.42) split into (2×2) matrices:

$$\begin{pmatrix} \tilde{\epsilon}_k & \Delta \\ -\Delta & -\tilde{\epsilon}_k \end{pmatrix} \begin{pmatrix} u_k \\ v_k \end{pmatrix} = \begin{pmatrix} u_k \\ v_k \end{pmatrix} \cdot E_k \tag{7.71}$$

with the gap parameter (7.75) $\Delta = p_0$, which have the BCS solutions (6.59).

To summarize, then, the complete solution of the HFB equations in the pairing-plus-quadrupole model corresponds to a Nilsson diagonalization with variable deformation parameters q_0 and q_2, a subsequent BCS calculation with constant gap parameter $\Delta = p_0$, and a minimization of the total energy

$$E(q_0, q_2, \Delta) = \langle \Phi | H | \Phi \rangle = \sum_k \tilde{\epsilon}_k v_k^2 + \frac{1}{2\chi}(q_0^2 + 2q_2^2) - \frac{\Delta^2}{G} \tag{7.72}$$

with respect to parameters q_0, q_2, and Δ. The term $(1/2\chi) \cdot (q_0^2 + 2q_2^2)$ has to be added, since in summing up single-particle energies we have counted the (negative) interaction energy twice [Eq. (5.40)].

This force explains the Nilsson model with BCS occupation probabilities very nicely and by a suitable choice of the constants G and χ one can therefore reproduce all its results. We have to emphasize, however, that it is only a model constructed for certain phenomena, namely quadrupole deformations and $I = 0$ pairing correlations and the interplay between these degrees of freedom. For these phenomena it contains all the gross features of a more realistic approach.

7.5 Applications of the HFB Theory for Ground State Properties

There are many HFB studies on *light nuclei* (see, for instance, [WFS 71, BKD 73]). A variety of interactions have been used. In many cases it is necessary to use projection of angular momentum and particle number (for instance [GGF 73] or [GGA 74]; see also Chap. 11). We will not go into the details, but mention here the so-called *proton–neutron pairing* (for a review, see [Go 79]).

In light nuclei protons and neutrons occupy the same levels. We can therefore construct not only *pp*- and *nn*-pairs with a large spacial overlap of their density distributions, but also *pn*-pairs [Go 64]. We thus have two types of pairing

correlations: $T=0$ pairing for *pn*-pairs only and $T=1$ pairing for *pp*-, *nn*- and *pn*-pairs. In heavy nuclei, only $T=1$ pairing is important. The full HFB equations (7.42) with a general force include all these types of pairing at the same time. This, however, requires complex potentials Γ and Δ and complex HFB coefficients U and V [GSB 70]. In $N=Z$ nuclei, not only is there time reversal, but also charge conjugation invariance. In this case it can be shown analytically [Go 72] that there are three possibilities for each particle. The proton can be paired with (i) a particle of the same kind in the time reversed state ($p\bar{p}$), (ii) a particle of different kind in the time reversed state ($p\bar{n}$), and (iii) a different particle in the same state (pn). The last possibility is not excluded by the Pauli principle; it does, however, violate the axial symmetry because it mixes states with different K-values ($+K$ and $-K$). The different modes have three different gap parameters, whose absolute squares add up to the square of the total gap.

Several authors have investigated different types of such pairing correlations in light nuclei, with many different kinds of forces [GK 65, CCJ 65+66, CG 67, GSG 68, BGG 69, WFS 70, WFS 71]. These calculations have demonstrated that $T=0$ pairing is significant in determining the equilibrium shapes. We find that for $N=Z$ nuclei, $T=0$ pairing is most important. For a small neutron excess, we have a competition between $T=0$ and $T=1$ pairing. For heavy nuclei with large neutron excess, where the protons and neutrons at the Fermi surface occupy different shells, $T=0$ pairing is no longer important and there remains the usual $T=1$ pairing in $p\bar{p}$ and $n\bar{n}$ pairs as discussed in Chapter 6.

To discuss further applications, we will restrict ourselves to a few examples in *heavy deformed nuclei*. According to the amount of phenomenology used in such calculations, we must distinguish among the following.

(a) Calculations in a Model Space. In this case, one starts with a finite configuration space and spherical single-particle energies. A phenomenological residual interaction is introduced which produces deformations and pairing correlations. Such theories are never able to give correct ground state energies.

The simplest model of this type is the pairing-plus-quadrupole model discussed in Section 7.4. By adjusting the force parameters it is possible to reproduce the deformation and the gap parameter in this way.

In the next step, Dietrich, Mang, and Pradal [DMP 66] used a more realistic residual interaction with finite range and spin–isospin exchange. They applied two methods to determine a HFB wave function $|\Phi\rangle$. In the first case they used Nilsson + BCS wave functions $|\Phi(\beta, \Delta)\rangle$ depending on the two parameters β and Δ and minimized the energy $\langle\Phi(\beta, \Delta)|H|\Phi(\beta, \Delta)\rangle$ with respect to these two parameters. In Fig. 7.1 we show the "energy surface" as a function of β. We see two minima for prolate and oblate deformations, which are rather similar in energy.

In the second case, they solved the full HFB equations $H^{20}=0$. Since this procedure allows for more general wave functions $|\Phi\rangle$, the resulting energies are lower (given by horizontal lines in Fig. 7.1). The energy difference is not very large in either case and a closer inspection of the wave function shows that the full HFB wave functions are rather well approximated by the "Nilsson+BCS" functions.

Although deformation parameters calculated by the second method give the trend quite nicely, they do not agree completely with experimental data, because of the restricted configuration space. On taking into account a contribution of the inert core, one finds full agreement.

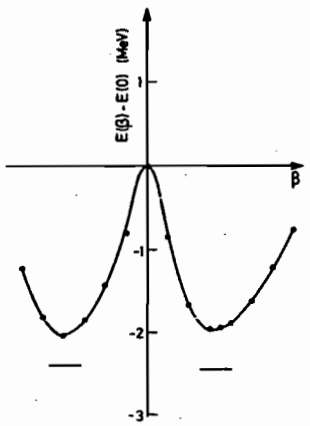


Figure 7.1. Binding energy as a function of deformation for ^{152}Sm. The lines beyond the minima are the solutions of the self-consistent calculations. (From [DMP 66].)

(b) Calculations with Skyrme Forces.* We saw in Section 5.6.2 that HF theory with density dependent Skyrme forces is able to reproduce the ground state properties of spherical nuclei. They have therefore also been applied to deformed cases [Va 73]. As discussed in Section 7.4, there is no a priori reason to use the same force for the calculation of Γ and Δ. The first step is therefore to use the pure pairing force (4.140) for the calculation of Δ. In the case of time reversal invariance, we again find (as in the pairing plus quadrupole model) that the canonical basis agrees with the HF basis, in which h is diagonal. It is thus sufficient to diagonalize h at each step of the iteration. The only difference to a normal HF calculation is that we have to take into account the pairing energy and occupation probabilities v_k^2 in the calculation of the densities $\rho, \tau, \mathbf{J}$. Instead of (5.84), we thus get, for instance,

$$\rho(\mathbf{r}) = \sum_{ist}^{\infty} v_i^2 |\varphi_i(\mathbf{r}, s, t)|^2, \tag{7.73}$$

with

$$v_i^2 = \frac{1}{2}\left(1 - \frac{\epsilon_i - \lambda}{\sqrt{(\epsilon_i - \lambda)^2 + \Delta^2}}\right). \tag{7.74}$$

Here λ is again fixed by the particle number condition. In principle, Δ should be determined by the gap equation (6.60). Vautherin went the opposite way and used a fixed gap Δ which he took from experiment and calculated the corresponding force constant using Eq. (6.60). The great difference from the HF calculations in spherical nuclei comes from the fact that in deformed nuclei the differential equation (5.100) is no longer separable in a radial coordinate r and angles θ, ϕ. It is now a partial differential equation. Vautherin has solved it by a diagonalization in an anisotropic harmonic oscillator basis. Only the matrix elements are calculated by a Gaussian integration in coordinate space. Axial symmetry allows a decompo-

*Similar calculations have been performed with the Negele force discussed in Section 5.6.1.2. [NR 77].

sition into subspaces with good quantum numbers Ω. Nevertheless, these calculations require a large computational effort.*

The results show a satisfactory agreement with the experimental binding energies and radii. In particular one can reproduce the measured quadrupole and hexadecupole moments of the deformed heavy nuclei [FQV 73].

(c) Full HFB Calculations. As we saw in Chapter 6, the pairing correlations depend not only on the properties of the effective pairing force, but also on the single-particle level density. A full HFB calculation, in which the self-consistent field and the pairing field are derived from the same force, is therefore only meaningful if the HF field yields a reasonable single-particle spectrum. As discussed in Chapter 5, one needs density dependent effective forces for such a task. On the other hand the present versions of the Skyrme force are not useful for complete HFB calculation because of their δ-character. Gogny [Go 73, 75b, DGG 75, DG 80] therefore introduced a new density dependent force (see Sec. 4.4.4) with a finite range part. The Gaussian form of this force allows a fast calculation of the matrix elements in a harmonic oscillator basis [Go 75a].

In the derivation of the HFB equations, we have not used density dependent forces until now. In analogy to Section 5.6 in such a case we have to calculate the energy as a functional of the densities ρ and κ [we use the conventions Tr_1 and Tr_2 defined in Eq. (E.19)],

$$E'(\rho,\kappa)=\langle\Phi|T+V(\rho)-\lambda\hat{N}|\Phi\rangle=\mathrm{Tr}\,(t-\lambda)\rho+\tfrac{1}{2}\mathrm{Tr}_1\mathrm{Tr}_1\rho\bar{v}\rho+\tfrac{1}{4}\mathrm{Tr}_2\mathrm{Tr}_2\kappa^*\bar{v}\kappa, \tag{7.75}$$

and derive the HFB equations from the variational principle. This is most easily achieved by working with the supermatrix $\mathcal{R}$ (7.27). We use $\rho_{ii}, \rho_{ik}, \rho_{ik}^*, \kappa_{ik}, \kappa_{ik}^*$ for $(i<k)$ as independent variables and impose the relations (7.28) as subsidiary conditions by a matrix of Lagrange parameters (this procedure could have also been adopted to derive the ordinary HF Eqs., Chap. 5)

$$\delta\big(E'(\rho,\kappa)-\mathrm{Tr}(\Lambda(\mathcal{R}^2-\mathcal{R}))\big)=0. \tag{7.76}$$

We find

$$\delta E'=\mathrm{Tr}(\mathcal{H}\cdot\delta\mathcal{R}) \tag{7.77}$$

with

$$\delta\mathcal{R}=\begin{pmatrix}\delta\rho & \delta\kappa\\ -\delta\kappa^* & -\delta\rho^*\end{pmatrix},\qquad \mathcal{H}=\begin{pmatrix}h & \Delta\\ -\Delta^* & -h^*\end{pmatrix} \tag{7.78}$$

and

$$h_{ik}=\frac{\delta E'}{\delta\rho_{ki}},\qquad \Delta_{ik}=\frac{\delta E'}{\delta\kappa_{ik}^*} \tag{7.79}$$

in agreement with Eq. (7.41) in the case of density independent forces.

From (7.76), it follows that

$$\mathrm{Tr}((\mathcal{H}-\Lambda\mathcal{R}-\mathcal{R}\Lambda+\Lambda)\delta\mathcal{R})=0. \tag{7.80}$$

Since $\delta\mathcal{R}$ is an independent variation, this is equivalent to

$$\mathcal{H}=\Lambda\mathcal{R}+\mathcal{R}\Lambda-\Lambda. \tag{7.81}$$

*For a solution of deformed HF-equations in the coordinate space, see [HN 77].

Λ is eliminated by the condition $\mathcal{R}^2=\mathcal{R}$ with the result:

$$[\mathcal{H}, \mathcal{R}]=0. \tag{7.82}$$

This is exactly equation (7.43). It is equivalent to Eq. (7.36) and can be solved in the same way as the solution of the HFB equations (7.42). However, in cases where the interaction $v(1, 2)$ depends on ρ, we get additional terms contributing to the self-consistent field Γ. From Eqs. (7.75) and (7.79) we see that they have the form:

$$\tfrac{1}{4}\frac{\partial}{\partial \rho_{ki}}\mathrm{Tr}_2\mathrm{Tr}_2(\kappa^*\bar{v}[\rho]\kappa). \tag{7.83}$$

By adjusting the force parameters in a suitable way (Table 4.4), Gogny was able to reproduce the properties of spherical as well as deformed nuclei. In particular, he got the proper deformations and gap parameters (see Table 7.1).

Table 7.1 Comparison between experimental and calculated values for the binding energy E, the charge quadrupole moments Q, and the gap parameters Δ for protons and neutrons (from [Go 75b])

	E[MeV]		Q[fm^2]		Δ_p[MeV]		Δ_n[MeV]	
	HFB	EXP	HFB	EXP	HFB	EXP	HFB	EXP
^{150}Sm	1235.8	1239.3	383	367	1.24	1.43	1.13	1.13
^{152}Sm	1249.0	1253.1	572	590	1.14	1.08	1.09	1.24
^{166}Er	1348.1	1351.6	770	770	1.10	0.97	1.02	1.00
^{184}W	1468.2	1472.9	631	627	0.90	0.86	0.86	0.73

The binding energies should still be corrected for the spurious contribution due to the violation of the rotational symmetry and particle number (see Chap. 11). These corrections can be of the order 5 MeV [FQV 73].

Summarizing these results we find that HFB theory with density dependent forces is able to reproduce all the essential ground state properties of spherical and deformed nuclei over the whole range of the periodic table. In the next section we will see how this theory can be extended to a microscopic calculation of energy surfaces for the fission process.

7.6 Constrained Hartree–Fock Theory (CHF)

Unrestricted HF and HFB calculations give only one point on the energy surface, namely the local minimum. We are often interested in other points on the surface, too: We can impose certain subsidiary conditions and ask for a minimum of the energy on the hypersurface defined by them. Sometimes we want to calculate the energy surface as a function of one (or several) collective parameters q, such as quadrupole or hexadecupole deformations. As we saw in Chapter 1, this is a first step toward a derivation of a collective Hamiltonian, as we use it in descriptions of collective vibrations or of the fission process. We will discuss this in more detail in Chapter 10 in connection with the generator coordinate method.

In all these cases, we are interested in a wave function $|\Phi(q)\rangle$, which minimizes the total energy under the *constraint* that a certain single-particle operator $\hat{Q}$ has a fixed expectation value

$$q = \langle\Phi|\hat{Q}|\Phi\rangle. \tag{7.84}$$

The simplest method for solving this problem is adding the condition (7.84) with a Lagrange multiplier λ to the Hamiltonian H and minimizing

$$\langle H'\rangle = \langle H\rangle - \lambda\langle\hat{Q}\rangle. \tag{7.85}$$

After the solution of this problem, λ is determined by the condition (7.84). This method is also called variation with a *linear constraint*. It corresponds to the investigation of the many-body system in an external field $\hat{Q}$ which deforms it in a proper way.

This method has already been applied in the BCS theory (Sec. 6.3.2) to get wave functions with the right expectation value of the particle number. In analogy to Eq. (6.41), we find that the Lagrange multiplier λ is the derivative of the energy $E = \langle\Phi|H|\Phi\rangle$ with respect to q

$$\lambda = \frac{dE}{dq}. \tag{7.86}$$

As we see from Fig. 7.2, the method of linear constraint means that we have to draw the tangent to the energy surface $E(q)$ at the point $q = q_0$ and use this line as the x-axis of a new coordinate system rotated by the angle α with $\tan\alpha = \lambda$. The y-axis in the new frame corresponds to $\langle H'\rangle$. An unrestricted variation on $\langle H'\rangle$ gives the extremum of the energy surface in the rotated system.

It is evident that this procedure can only work as long as the function $E(q)$ has a positive second derivative. In cases where the curve is concave downwards—as, for instance, in the neighborhood of a saddle point on the fission path or in the backbending region (Fig. 3.4)—this method will no longer work. In such cases, the energy in the rotated frame has a maximum at the point q_1 which will never be a stable solution of the HF equations.

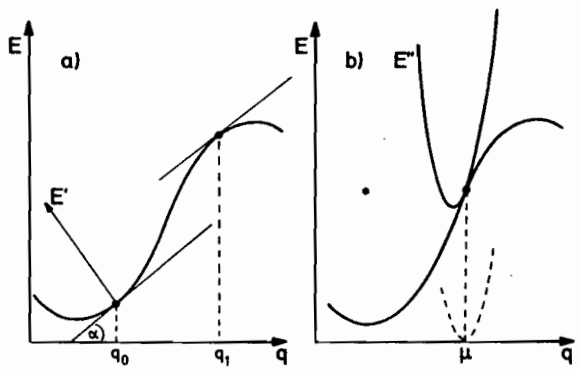


Figure 7.2. Schematic representation of an energy surface showing the methods of linear (a) and (b) quadratic constraint.

We also recognize that in the vicinity of inflection points there are always several points with parallel tangents (that is, with the same λ, but different values of q), that is, q is no longer a unique function of the Lagrange parameter (see Fig. 3.6).

There are several ways to avoid this problem and to find the wave function for such points in the energy surface.

(i) The gradient method (as discussed in Sec. 7.3) allows us to follow the line of steepest descent perpendicular to the gradient with respect to $\hat{Q}$. Therefore, for each value of q we find the corresponding minimum in the energy.

(ii) The method of quadratic constraint [GLW 70, FQK 73] uses an unrestricted variation of the function

$$E'' = \langle H \rangle + \tfrac{1}{2} C(\langle \hat{Q} \rangle - \mu)^2. \tag{7.87}$$

As in Fig. 7.2b, here we add a parabola with the symmetry axis $q = \mu$ to the energy surface. It produces a new surface E'' with an extremum in the vicinity of μ. If its width is sufficiently small ($C > |d^2E/dq^2|$), this is always a real minimum.

The variation of E'' gives

$$\delta\langle H \rangle - C(\mu - \langle \hat{Q} \rangle)\,\delta\langle \hat{Q} \rangle = 0. \tag{7.88}$$

This means the quadratic constraint is equivalent to a linear constraint with $\lambda = C(\mu - \langle \hat{Q} \rangle)$. In both cases, we obtain the *same* energy surface. This is true for any constraint that is a function of $\langle \hat{Q} \rangle$. With the quadratic constraint, the effective value of λ is automatically changed during the iteration.

(iii) A third method [BW 71] uses a linear constraint but changes the value of λ at each step of the iteration in such a way that the condition (7.84) is fulfilled. This method is applied in BCS calculations with a constraint on the particle number. However, it requires an additional iteration at each step.

Constrained Hartree–Fock theory (CHF) has been used to calculate the energy surfaces of nuclei in the vicinity of the ground state [BKS 67, CLF 73, CS 73, FQK 73, GG 78] and for fission processes [FQV 74, KCS 74, BFV 74].

In the latter case usually the question has to be faced of which constraining operator $\hat{Q}$ should be used. It is easy to show that we can find any product wave function $|\Phi\rangle$ by CHF calculation using a suitable constraining operator $\hat{Q}_\Phi$. We have only to use an operator whose two-quasi-particle part Q_Φ^{20} is defined as a multiple of H^{20}. We then get

$$H^{20} - \lambda Q^{20} = 0, \tag{7.89}$$

that is, $|\Phi\rangle$ as a solution of the corresponding CHF equation. Of course, this is not of much practical use for a calculation of $|\Phi\rangle$, but it shows that there is a large variety of constraining operators that produce the same wave function and are different in their Q^{11}-matrix elements.

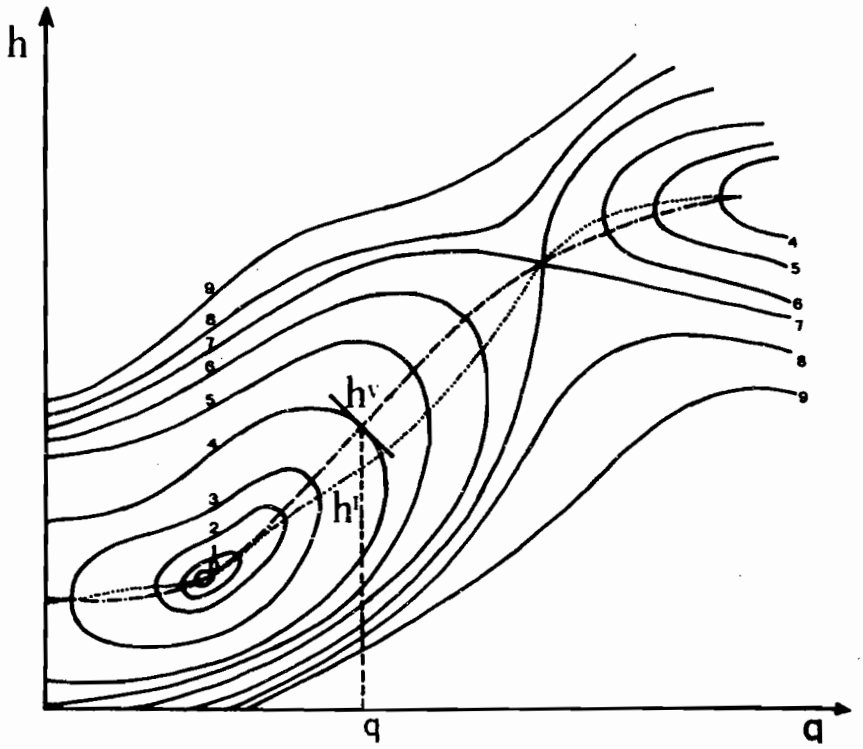


Figure 7.3. Example of an energy surface in the (q, h) plane showing that we do not follow the fission valley exactly with a constraint only on q. (From [FQK 73].)

On the other hand, a constraint on one deformation (for instance, the quadrupole deformation $q = \langle \Phi | Q_{20} | \Phi \rangle$) does not mean that another deformation (for instance, the hexapole deformation $h = \langle \Phi | Q_{40} | \Phi \rangle$) stays constant. This is shown schematically in Fig. 7.3. As long as we use a constraint on q, we follow the line $h^1(q) = \langle \Phi(q) | Q_{40} | \Phi(q) \rangle$. It connects the points of a vertical tangent on each equipotential surface, because E has to be a minimum for constant q with respect to all other degrees of freedom, which means, in particular, $dE/dh \mid_{q=\text{const}} = 0$. This line $h^1(q)$ is, in general, different from the bottom of the fission valley, which corresponds to a line of steepest descent $h^0(q)$. Both lines go through the stationary points of the surface, such as the ground state minimum or the saddle point. Both lines are very close if the fission valley is parallel to the q axis, but they can deviate quite drastically if the valley is parallel to the h-axis. Nevertheless, we recognize an enormous advantage of CHF calculations compared to the liquid drop and Strutinsky-type calculations (see Sec. 2.9): In the CHF calculation we need only follow a one-parameter line and obtain a wave function which, in many cases, follows quite closely the bottom of the fission valley, whereas in the other methods we must calculate multidimensional energy surfaces.

We should certainly optimize the choice of the constraining operator in CHF calculation. In Chap. 10 we shall study a method for achieving this.

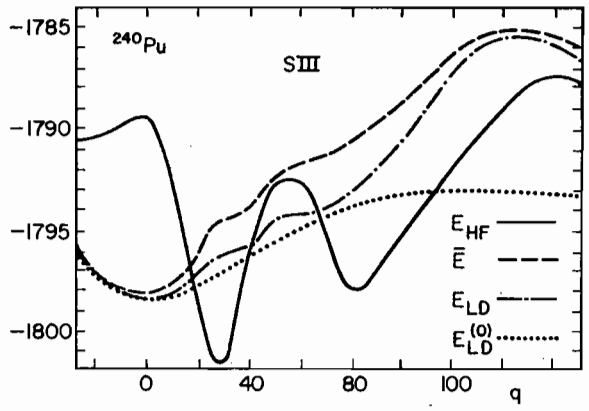


Figure 7.4. Deformation energy curves of ^{240}Pu obtained with the Skyrme III interaction as a function of the mass quadrupole moment q. The details are explained in the text. (From [BQ 75a].)

Figure 7.4 shows the deformation energy for the nucleus ^{240}Pu calculated with a Skyrme force with a constraint on the mass quadrupole moment q (full line [FQV 74]). Pairing correlations are taken into account by a simple pairing force, whose constant G is chosen to be proportional to the surface. The calculation reproduces the ground state minimum and a second minimum. The heights of the barriers do not agree quantitatively with experimental data, but we must bear in mind that axial symmetry and parity are conserved in the calculation, whereas there are indications that the actual fission path violates axial symmetry at the inner and parity at the outer saddle.

To compare these results with a liquid drop calculation, a Strutinski averaged HF energy (dashed line) is calculated and shown in Fig. 7.4. It shows small wiggles. These are, however, not caused by shell effects but are also present in the liquid drop energy which corresponds to the same deformation q and h (dashed-dotted line), because the curve intersects with side valleys. The liquid drop energy at the bottom of the fission valley (dotted line) is very smooth.

Figure 7.5 shows density distributions for the same nucleus at different deformations.

Several authors [FS 76, BSV 76] have pointed out that CHF calculations of this type with constraining operators with unbounded spectra (like $Q \sim r^2 Y_{20}$, which diverges for $r \to \infty$) can give very spurious results if wave functions $|\Phi\rangle$ are allowed to spread out very far in space. All the practical applications of the CHF method, however, have been carried out in finite configuration spaces, which ensures that the wave functions $|\Phi\rangle$ are somehow concentrated in the region of the nucleus.

In the choice of the configuration space, therefore, we are limited from two sides. To get convergence in the binding energy, a large configuration space is often needed, but on the other hand we should not be able to represent the spurious solutions mentioned above. It seems that the CHF calculations made up to the present time stay within these limits.

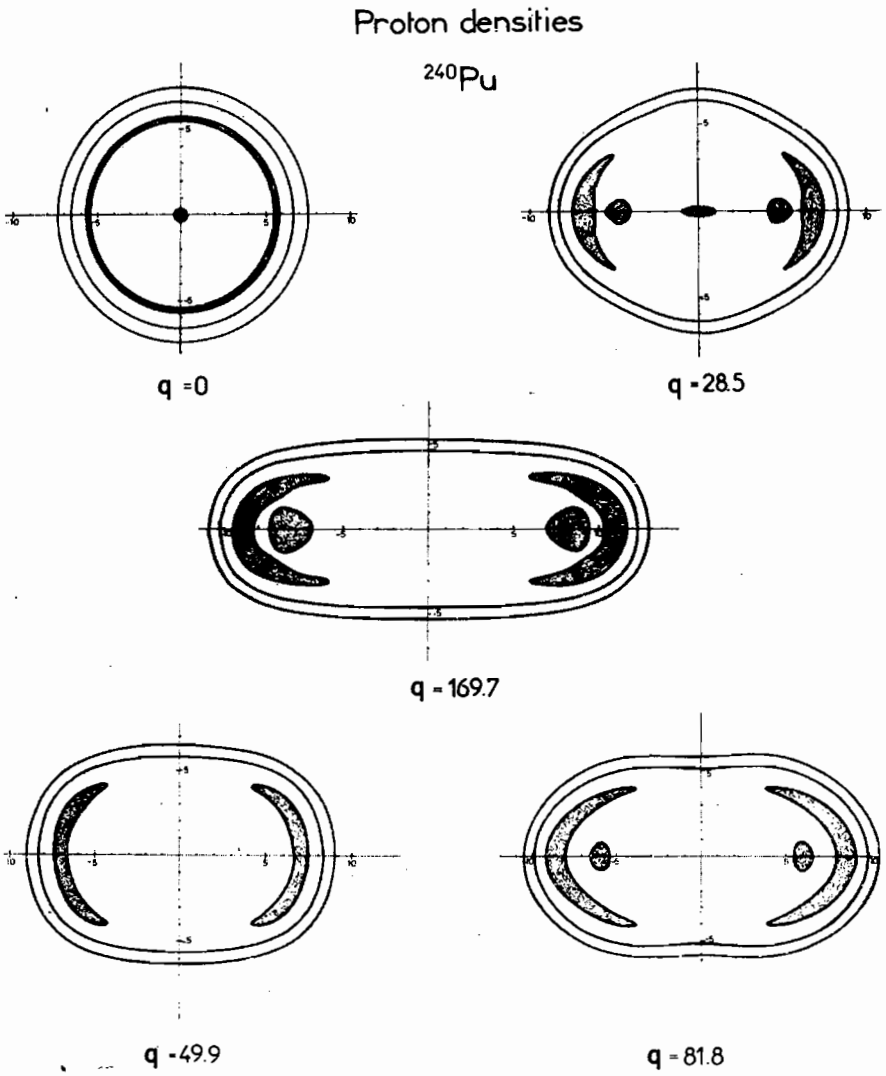


Figure 7.5. Density distributions of ^{240}Pu at different deformations $q=\langle Q_{20}\rangle$. (From [FQV 74].)

A different possible way of keeping the nuclear wave function concentrated is to introduce a second constraint [GL 77]:

$$\hat{H}' = \hat{H} - \lambda\hat{Q} + \mu\hat{Q}^2. \tag{7.90}$$

The Lagrange parameter μ would allow us to vary the $\langle\Delta\hat{Q}^2\rangle$ fluctuation in the wave function.

7.7 HFB Theory in the Rotating Frame (SCC)

In Chapter 3 we discussed the cranking model within the framework of the pure single-particle shell model. It gave us the possibility of calculating the moment of inertia microscopically. It is evident how to generalize this

theory to the self-consistent many-body case: we must vary the expectation value of the many-body Hamiltonian H with the subsidiary condition that the angular momentum operator J_x has a certain expectation value. This gives the equation (see also Sec. 11.4)

$$\delta\langle\Phi|H-\omega J_x|\Phi\rangle=0. \tag{7.91}$$

The cranking frequency ω depends on the angular momentum I and is determined by

$$\langle\Phi_\omega|J_x|\Phi_\omega\rangle=\sqrt{I\cdot(I+1)}\ . \tag{7.92}$$

In Chapter 11 we will see that the solution $|\Phi_{\omega(I)}\rangle$ of this problem must be understood as an internal wave function, from which the eigenfunction of $\mathbf{J}^2$ can be obtained in the laboratory system by a projection onto good angular momentum.

Within the HFB theory, we restrict the set of variational functions $|\Phi\rangle$ to generalized product functions of the HFB type, and we are then left with a constrained HFB problem. In analogy to the cranking model in the single-particle case, this method is called the *self-consistent cranking model* (SCC). We have already discussed in great detail the problems of the constrained HF method in the last section. In the backbending region (see Chap. 3 and [CMR 75]), the solution of (7.91) corresponds to a saddle point in the energy surface with constant ω. Therefore, we have to apply the gradient method or a quadratic constraint to obtain the solution on the backward-going branch.

In contrast to the applications of the HFB theory discussed so far, the time reversal symmetry of the wave function $|\Phi\rangle$ is broken by the operator ωJ_x. Therefore, we do not know a priori the conjugate states in the canonical basis and thus we must solve the full HFB problem in the rotating frame [RBM 70] viz:

$$\begin{pmatrix} h-\omega j_x & \Delta \\ -\Delta^* & -h^*+\omega j_x^* \end{pmatrix}\begin{pmatrix} U_k \\ V_k \end{pmatrix}=\begin{pmatrix} U_k \\ V_k \end{pmatrix}E_k. \tag{7.93}$$

Using the symmetry $e^{i\pi J_x}$, which is conserved by the cranking Hamiltonian, the matrices ρ, κ, Γ, and Δ take the form (7.67). Since the time reversal symmetry T is violated, we have, instead of Eq. (7.69),

$$\rho_2\neq\rho_1^*;\qquad \Gamma_2\neq\Gamma_1^*;\qquad D_2\neq D_1^*;\qquad \kappa_1^+\neq\kappa_1;\qquad \Delta_1^+\neq\Delta_1. \tag{7.94}$$

The densities and the potentials thus have a time-even and a time-odd part, as, for example,

$$\rho^{(+)}=\rho_1+\rho_2^*;\qquad \rho^{(-)}=\rho_1-\rho_2^*. \tag{7.95}$$

The quasi-particle energies E_k which, for $\omega=0$, are twofold degenerate (see Sec. 2.8), split up at finite cranking frequencies (Fig. 7.8d) and are no longer given by formula (7.51). According to the Bloch–Messiah theorem (7.7), however, there exists a canonical basis, which can be found by a diagonalization of the density ρ. As we can see from Eq. (7.50), the

occupation probabilities v_k^2 in this basis are given by

$$v_k^2 = \frac{1}{2}\left[1 - \frac{\epsilon_k - \omega j_{x_k}}{\sqrt{(\epsilon_k - \omega j_{x_k})^2 + \Delta_{k\bar{k}}^2}}\right], \tag{7.96}$$

with $\epsilon_k = \frac{1}{2}(h_{kk} + h_{\bar{k}\bar{k}})$ and $j_{x_k} = \frac{1}{2}(j_{x_{kk}} + j_{x_{\bar{k}\bar{k}}})$.

This shows that the diagonal matrix elements h_{kk}, $j_{x_{kk}}$, $h_{\bar{k}\bar{k}}$, $j_{x_{\bar{k}\bar{k}}}$, and $\Delta_{k\bar{k}}$ in the canonical basis are the crucial quantities that determine the occupation probabilities. The matrices h, j_x, and Δ themselves are in general not diagonal in this basis. One also notices that levels with large positive matrix elements j_{x_k} have a greater occupation probability than those with large negative values.

The self-consistent cranking model is useful for the description of the

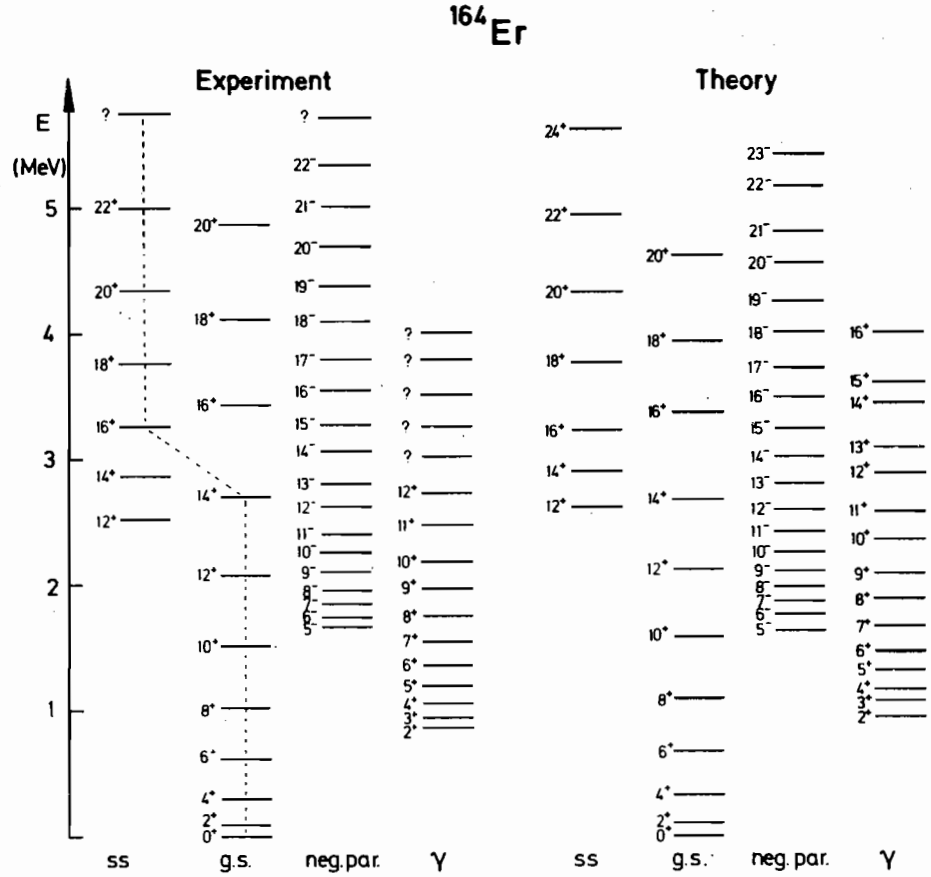


Figure 7.6. Spectra of rotational bands in ^{164}Er. Besides the ground state band (gs), we see a Stephens–Simon band (ss) of an aligned $i13/2$ neutron pair, a negative parity band, and a γ-band [KSM 78]. The yrast line (indicated by a dashed line) follows the lowest-energy levels for each I-value. The r.h.s. gives theoretical calculations within the SCC method [MSR 76, EMR 80a + b].

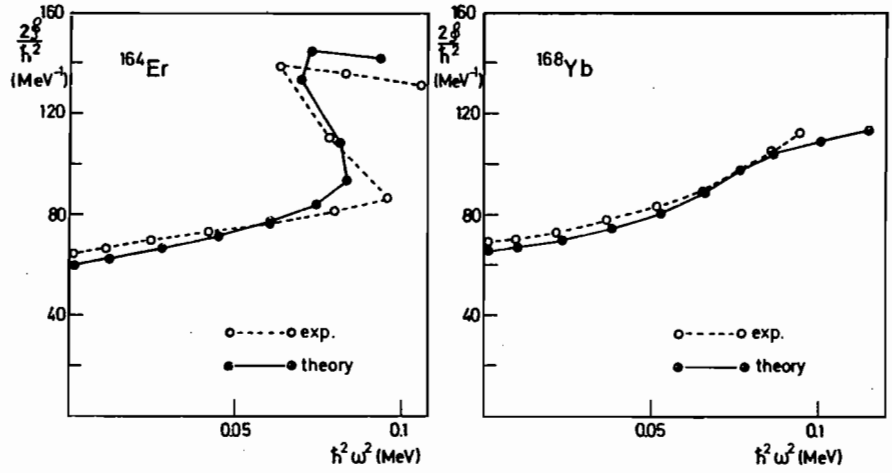


Figure 7.7. Backbending plots for the nuclei ^{164}Er and ^{168}Yb. Dashed lines represent experimental data and full lines show a calculation within the SCC theory [MSR 76]. In both cases, Eq. (3.3) was used to define the moment of inertia $\mathcal{I}$ and the angular frequency ω.

type of anomalies found in the high spin region, * which were discussed in Chapter 3. It is general enough to embrace a great variety of physical effects. To illustrate, we will discuss some of these effects for the nucleus ^{164}Er, which has been very well experimentally investigated in the last few years [LCS 76, KSM 78]. Several rotational bands have been found (see Fig. 7.6) which go up to rather high spin values.

A variation of Eq. (7.91) yields the lowest energy for each I-value. This is, by definition, the yrast line (dashed line in Fig. 7.6). As we see, it follows the ground state band only up to $I=14$. Between $I=14$ and $I=16$, we have a band-crossing, that is, we get backbending if we follow the yrast line (see Fig. 7.7). The crossing band has the structure of a two-quasi-particle band with two aligned $i_{13/2}$ neutrons. Therefore, the yrast line changes its character. It follows the ground state band up to $I=14$ and then corresponds to the aligned band for all I values $I \geqslant 16$. The calculation of levels along the yrast line within the self-consistent cranking model shows the following features.

(i) The Stretching Effect represents the possible change of the nuclear deformation during the rotation. It is described by the self-consistent potential Γ, which depends on the density and therefore on the cranking frequency. Figure 7.8a shows the effective nuclear deformations β and γ found in the SCC calculation for ^{164}Er. The change in the deformation β is rather small in this spin region. The same is true for many of the well-deformed rare earth nuclei. On the other hand, the γ-deformation

* Recently, the validity of the cranking model for the description of bandcrossings has become an object of discussion [Ha 76, 77, MG 78, GSF 78b, CDS 78].

goes up to $\gamma \sim 10°$, but this depends strongly on the particular nucleus, and other nuclei can be rather stiff against γ-deformations.

(ii) The Coriolis Antipairing Effect (CAP) means that the pairing correlations within the nucleus are weakened by the Coriolis force. In Chapter 6 we saw that the pairing correlations come mostly from nucleon pairs of opposite spin (Cooper pairs), which feel a strong interaction because of their large spatial overlap. Since the Coriolis force acts on both angular momenta in the opposite direction (Fig. 7.9), we should expect at least a partial alignment and a lowering of the corresponding pair energy Δ with increasing spin. This is observed in the calculations (Fig. 7.8b) and is responsible for the moderate increase of the moment of inertia at low spin

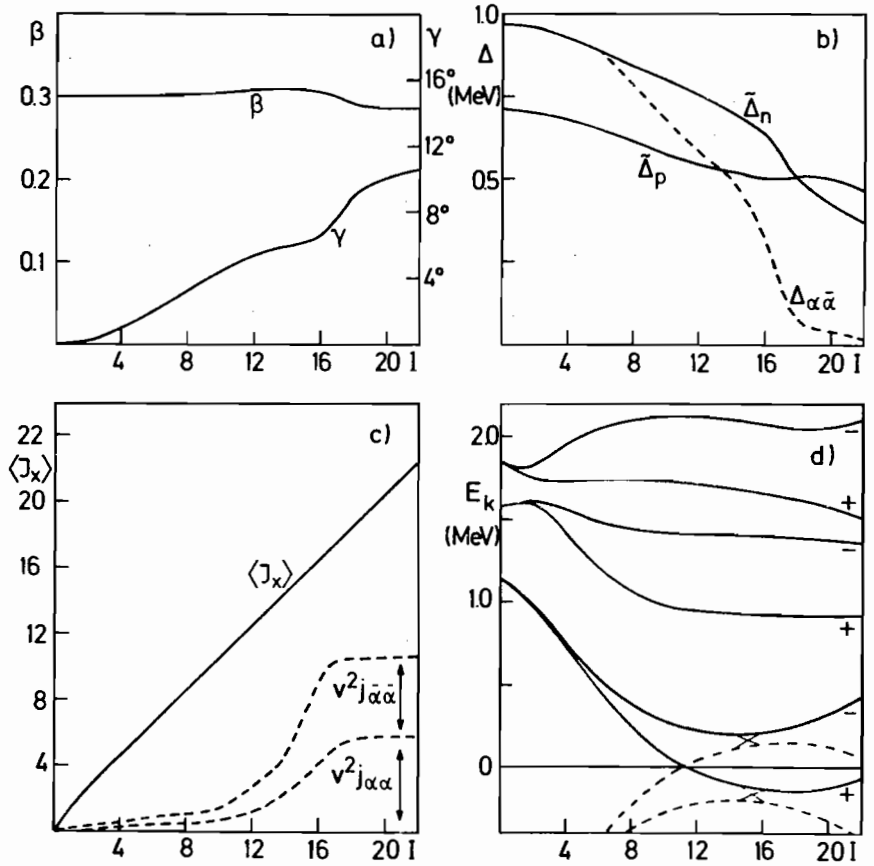


Figure 7.8. Calculated deformation parameters (a), gap parameters (b), expectation values of the angular momemtum J_x (c), and quasi-particle energies E_k in the rotating frame (d) as a function of the angular momentum for the case of the nucleus ^{164}Er. (From [MSR 76].) The value of the β-deformation is normalized to 0.3 for $I=0$. Part (b) shows the averaged gap parameters $\tilde{\Delta}_p$ and $\tilde{\Delta}_n$ in the canonical basis [Eq. (7.97)] for protons and neutrons and the gap parameter $\Delta_{\alpha\bar{\alpha}}$ of the aligned pair (dashed line). Further details are explained in the text.

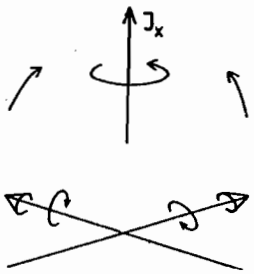


Figure 7.9. Schematic representation of the Coriolis antipairing effect.

values. As in a superconductor within a magnetic field, for a certain critical angular momentum we find a transition to a normal fluid state. This *nuclear Meissner effect* was predicted by Mottelson and Valatin [MV 60] and found in many calculations (for instance, [CV 66, RBM 70, Wa 70a, Ku 72, Fr 76]) at rather different spin values.

The backbending phenomenon has been assumed to be caused by such a phase transition [KS 73]. More realistic calculations, however, which allow decoupling processes of one or several pairs of nucleons, showed that the pairing gap Δ does not vanish completely in the backbending region (Fig. 7.8b, between $I=12$ and $I=16$). It is only reduced because of the alignment process. This corresponds to a transition to a two-quasi-particle band with the corresponding blocking effect (see Sec. 6.3.4). In Fig. 7.8b, we show the averaged gap

$$\tilde{\Delta}=\frac{1}{M'}\sum_{k>0}^{M'}\Delta_{k\bar{k}}\,. \tag{7.97}$$

($M'=M/2$ is the number of levels $k>0$ in the configuration space.) Even for a constant pairing force with the constant pairing matrix element Δ in the original basis, not all of the matrix elements $\Delta_{k\bar{k}}$ in the canonical basis are equal. In particular, the matrix element of the aligned pair $\Delta_{\alpha\bar{\alpha}}$ goes to zero (dashed line in Fig. 7.8b). This effect is most drastically pronounced for a pure pairing force, which acts only on $I=0$ pairs (see Sec. 4.4.7). Realistic calculations for very high spin states still have to be done. Model calculations and experimental data [BSC 75], however, indicate that the remaining pairing correlations still persist, sometimes up to rather high spin values ($I\sim 40$–$50\hbar$).

(iii) Alignment Processes. (Stephens–Simon Effect, [SS 72a]; see Sec. 3.2). Since HFB wave functions $|\Phi\rangle$ are products of the most general single-particle states (with corresponding occupation probabilities), they can represent the ground state band as well as states of partially or totally aligned two-quasi-particle bands. Therefore, they are the appropriate tools for investigating such decoupling processes microscopically. It is clear, however, that we must allow for a violation of axial symmetry to describe

such aligned states, which are a mixture of many different Ω values (see Sec. 3.3). This is no problem in a cranking calculation, since the Coriolis term ωj_x destroys this symmetry anyway.

The degree of alignment is measured by the matrix elements $j_{x_{\alpha\alpha}}$ and $j_{x_{\bar{\alpha}\bar{\alpha}}}$ of the two particles in the canonical basis. Figure 7.8c shows these values [multiplied by the corresponding occupation probability (7.96) which are, in fact, very close to 1] and the total expectation value of J_x. One notices that in the back-bending region more than the total increment in J_x comes from this pair. The contribution from the rest of the particles decreases, which corresponds to a decrease in the angular frequency ω in this region. For higher spin values, the contribution of the aligned pair is rather constant $5.8+4.8\simeq 10.6$, whereas a fully aligned $i_{13/2}$ pair would give $13/2+11/2=12$.

(iv) Finally, Fig. 7.8d shows some quasi-particle energies E_k in the rotating frame, which are the eigenvalues of Eq. (7.93) as a function of angular momentum. They are no longer twofold degenerate, as is the case with time reversal invariance. At large angular momentum it may happen that one of these energies goes to zero and becomes negative. Since the pairing potential does not vanish in this region, this phenomenon is called *gapless superconductivity** [GLS 67, Gr 74, CMR 75]. It does not mean, however, that there is an excited state with lower energy, because the excitation energy in the even system is always the sum of two quasi-particle energies. We have seen that for each eigenvalue E_k of Eq. (7.93) we also have an eigenvalue $-E_k$ (dashed lines). One of these "unphysical" eigenvalues becomes positive in the backbending region. However, we are not allowed to choose this positive eigenvalue, because (as discussed in Sec. 7.3.2) this would correspond to a transition to a wave function in a system with odd particle number. We also indicated in Fig. 7.8d (by the thin lines) the band-crossing phenomenon, which is connected with the transition from the ground state band to a two-quasi-particle band in the backbending region.

Similar features are found in all nuclei in the rare earth region.† There are, however, drastic differences in the backbending-plots (see Fig. 7.7), because this kind of plot is very sensitive to small changes in the level spacings of the spectra. In many cases such alignment processes do take place without the backbending phenomenon. As model calculations [CMR 75] show, whether backbending occurs depends on the underlying single-particle level density and on the residual interaction between the two crossing bands: small interaction and low level density give sudden alignment and therefore backbending. In other cases (for instance, ^{168}Yb) we have a smooth alignment process without backbending (see also footnote on p. 122).

* An analogous phenomenon occurs in metallic superconductors [AG 61].

† For applications of the SCC theory for light nuclei see [PM 76, NS 78, Zi 78].

The first calculations of this type were carried out by the Munich group [BMR 73] with a pairing-plus-quadrupole interaction in a restricted configuration space. In the meantime, the theory has been improved by particle number projection [DBM 73, FDG 76]. As discussed in Chapter 11, this seems to be necessary in regions in which the pairing correlations undergo drastic changes. The quantitative agreement with experiment is somewhat improved by such calculation, but nothing has changed in the qualitative picture discussed above. The same holds for the inclusion of quadrupole pairing forces [WF 78a].

Realistic forces have also been used, like the G-matrix in a restricted space derived from the Reid soft core potential [GVS 76, Go 76a] or forces of the Skyrme type [FMR 79]. Qualitatively similar results are found.

In Section 7.3.2 we saw that we can use the functions $|\Phi\rangle$ of the HFB type not only for the ground state of even nuclei and for states along the yrast line, but also for all kinds of quasi-particle states in even and odd nuclei. Therefore, we can also describe, in the SCC method, rotational bands in odd nuclei and general two-quasi-particle bands in even nuclei. In particular, we can study all types of alignment processes.

A more fundamental derivation of the cranking model in Chapter 11 shows that we should use the subsidiary condition

$$\langle J_x\rangle=\sqrt{I\cdot(I+1)-\langle J_z^2\rangle} \tag{7.98}$$

instead of Eq. (7.92). For the ground state band in even nuclei, which are to a good approximation axially symmetric, $\langle J_z^2\rangle$ is rather small. In such cases (7.98) is equivalent to (7.92). With this subsidiary condition very disturbed rotational bands in odd nuclei have been calculated [RMB 74, FDB 77] and been found in good agreement with the experimental spectra (Fig. 3.11). In particular, there is no need for any attenuation of the Coriolis interaction in this model [RM 74, Ri 77].

Figure 7.6 shows an example of a two-quasi-particle band with negative parity. Here one neutron sits in the $i_{13/2}$ shell and the other in negative parity shells. We again observe an alignment process of one $i_{13/2}$ neutron, which causes a small level spacing for the lower members of this band. Because of the parity selection rule, the corresponding wave functions are automatically orthogonal to the ground state. This is different in the calculation of the so-called "second" band, which causes the backbending phenomenon. It is an excited band which changes its character at the backbending point from the aligned two-quasi-particle band to the ground state band. This band has the same symmetries as the yrast band. Therefore, we have to apply the additional constraint that its wave functions are orthogonal to the corresponding wave functions of the yrast band [EMR 80a].

Figure 7.6 also shows a γ-band. As we shall see in the next chapter, such collective vibrations cannot be described by pure product wave functions. This does not, however, mean that the rotational band built upon such a vibration cannot be investigated by the cranking model. We need only use,

for $|\Phi\rangle$, wave functions of a more complicated structure (like TDA or RPA wave functions which are discussed in the next chapter; [Ho 68, CMR 75, Ma 77a, ERM 78, 80b]).

Let us finish this chapter with the following concluding remarks: HFB theory is a method which includes as many correlations as possible in a generalized static single-particle picture. It therefore defines two self-consistently determined fields [the order parameters (see Chap. 11)], an averaged potential Γ for the *ph* correlations, and a pairing potential Δ for the *pp* correlations. Within this theory, we are able to reproduce the ground state properties of spherical and deformed nuclei. It also can be used for the description of excited states with wave functions of generalized product type, such as pure quasi-particle excitations or rotational states in well-deformed nuclei. In the latter case a constraint on the average angular momentum must be used. We should keep in mind, however, that deformed wave functions have to be understood as intrinsic wave functions, from which we can get the wave function in the laboratory system by a projection onto eigenstates of angular momentum (see Sec. 11.4).

In addition, the HFB theory can be used to provide an optimal basis for more complicated methods to treat further correlations. This will be done in following chapters.